





Al-Balqa' Applied University

Faculty of Engineering Technology

Mechatronics Engineering Department

Process control Lab

Ex. 1

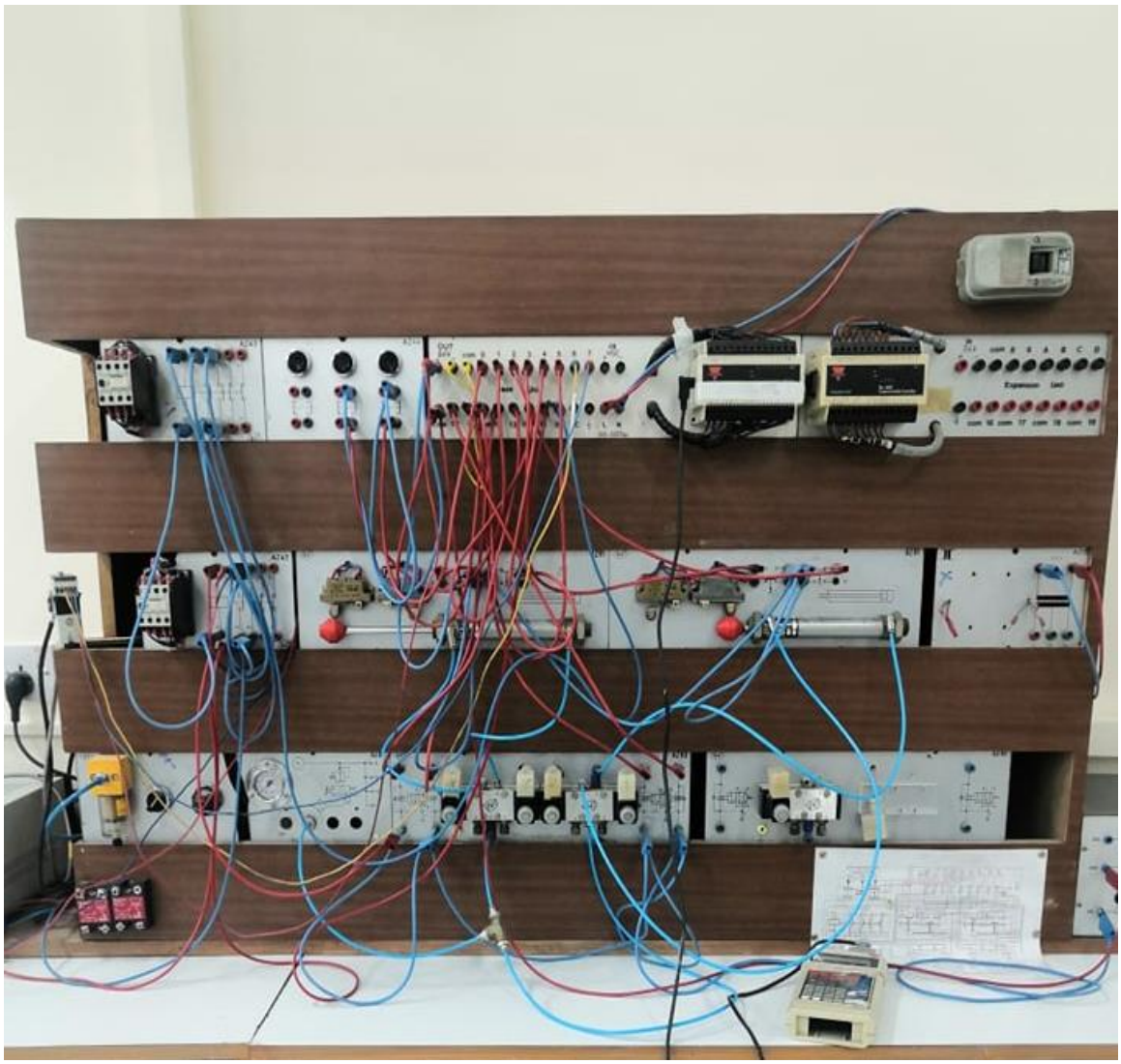
Control the liquid level

Using PLC (ELC 2001)

Student's Name:

Supervisor:

Date:



❖ Objective:

Training the student to develop ladder diagrams or logical implementation programs to control the level of liquids in tanks

❖ Necessary Equipment:

1. Programmable controller ELC 2001, can be programmed using a special part to do so (Program loader) the program can be put as a ladder diagram or instruction list.
2. Two plastic tanks, the second contains them three connecting electrodes.
3. Level regulator.
4. Electric pump by direct current at voltage 24 V DC.
5. Electrical relay type (RL/2).
6. Two contactors single phase AC voltage.
7. Power supply 24V DC and power supply 230 V AC.

❖ Introduction:



Fig (1-1)

A. Functional specifications:

- **I/O Points:**

Input Points: All points are allocated to 8 bit registers (P000 to P008) which means that every point address represents one single bit in such a register.
Name of register is P01

P00	0	1	2	3	4	5	6	7

Address every point by location that it's run for example:

1		3		6			

P001, P003, P006

Output Points: All points are allocated to 6 bit registers (P010 to P015) which means that every point address represents one single bit in such a register.
Name of register is P01

P01	0	1	2	3	4	5

Address every point by location that it's run for example:

P01	0		2			5

P010, P012, P015

- **Timer:**

Time base	100 Ms	(T000 to T191)
Time base	10 Ms	(T192 to T255)
Functions	5	(ON, OFF, Integr, monost, re-trig)

B. Manual Programmer ELC2000PL:

Parts Description:

1	Handle.
2	Jack Connector for External 5VDC Power Supply (ELC2000SV).
3	DB9 Connector for PLC Connection Cable (ELC200LCAB2).
4	Mode Keys with PLC Status Indicator LED's.
5	Display Contrast Adjustment.
6	Programming Command keys.
7	Clear and Enter Keys.
8	Slot for EPROM Write Adapter (ELC2000ELA).
9	Numeric Keys.
10	Register Type Selection Keys.
11	Function and Instruction Keys 2 Line 16 Character.
12	Dot Matrix LCD Display.

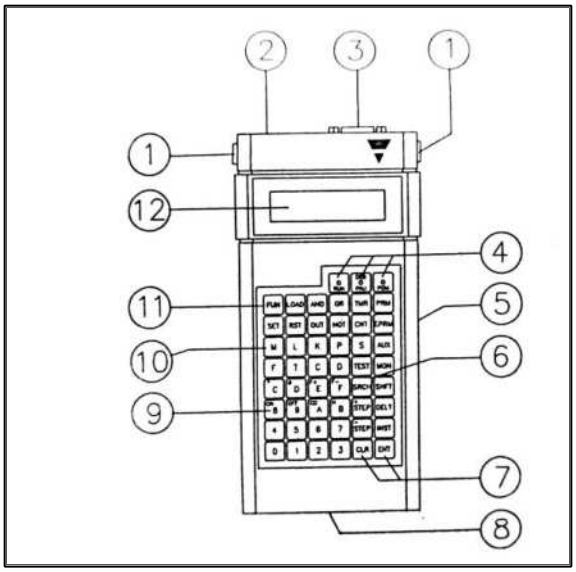


Fig (1-2)

C. Key function for PLC ELC 2001:

Key Type	Name	Function
Mode Key	RUN	Change PLC status to RUN (Application program is executed)
	PAU	Change PLC status from RUN condition to pause mode, or from programming mode to debug mode.
	PGM	Change PLC status to programming mode.
	+STEP	Step through application program (forward) or display next higher point or register-address during monitoring.
	-STEP	Step through application program (back) or display next lower point or register-address during monitoring.
	DELT	Delete one step of application program. (Delete enter program with SHFT - DELT).
	INST	Insert a program step in the application program.
Clear and Enter Keys	CLR	Clear last input or clear display.(Show initial display with SHFT CLR)
	ENT	Accept and enter a program step or parameter.
Numeric Keys	0 to F	Input addresses , step numbers, variables parameters, etc.
Type Select Keys	P	Input/output point (P00 … P19) or I/O register (P0 … P1)
	M	Auxiliary relay (M000 … M63F) or Aux.reg (M0 … M63)
	T	Timers (T000 … T255).
	C	Counters (C000 … C255).
	LOAD	Used for LOAD, LOAD NOT, AND LOAD, OR LOAD.
	AND	Used for AND, AND NOT, AND LOAD.
	OR	Used for OR, OR NT, OR LOAD.
	NOT	Used for LOAD NOT, AND NOT, OR NOT, NOT.
	OUT	Sequence output.
	FUN	Used for application instruction, D, DNOT, MCS, CLR, END.
	TMR	Timers TON, TOFF, TMR, TMON, TRTG (Press key a number of times to select desired counter function).
	CNT	Counters CTU, CTD, CTUD and CTR (Press key a number of

		times to select desired counter function).

D. Liquid Level Electrode



Level electrode holder designed to hold 2 level electrodes into a tank, process vessel, sump or pit.

Designed to automate a number of liquid level applications including when combined with a compatible Level Controller such as is offer the switching On/Off of pumps, open/close valves, or to alert to low or high levels.

E. Level Regulator:

The Small level regulator is a float switch which allows electrical equipment to start and stop automatically (usually pumps, but also electric control valves, alarms, etc.)

When a prefixed level has been reached, the most essential characteristics of this device are that it is small, highly reliable, and with a high switching potential since it is equipped with micro-switches that have high commuting features

Level control relay for conductive liquids selectable for filling and emptying applications with adjustable sensitivity. The unit can be wired for three wire application to control levels between High and Low probe or two wire applications for alarm indication. Filling or emptying can be selected via front slide switch.

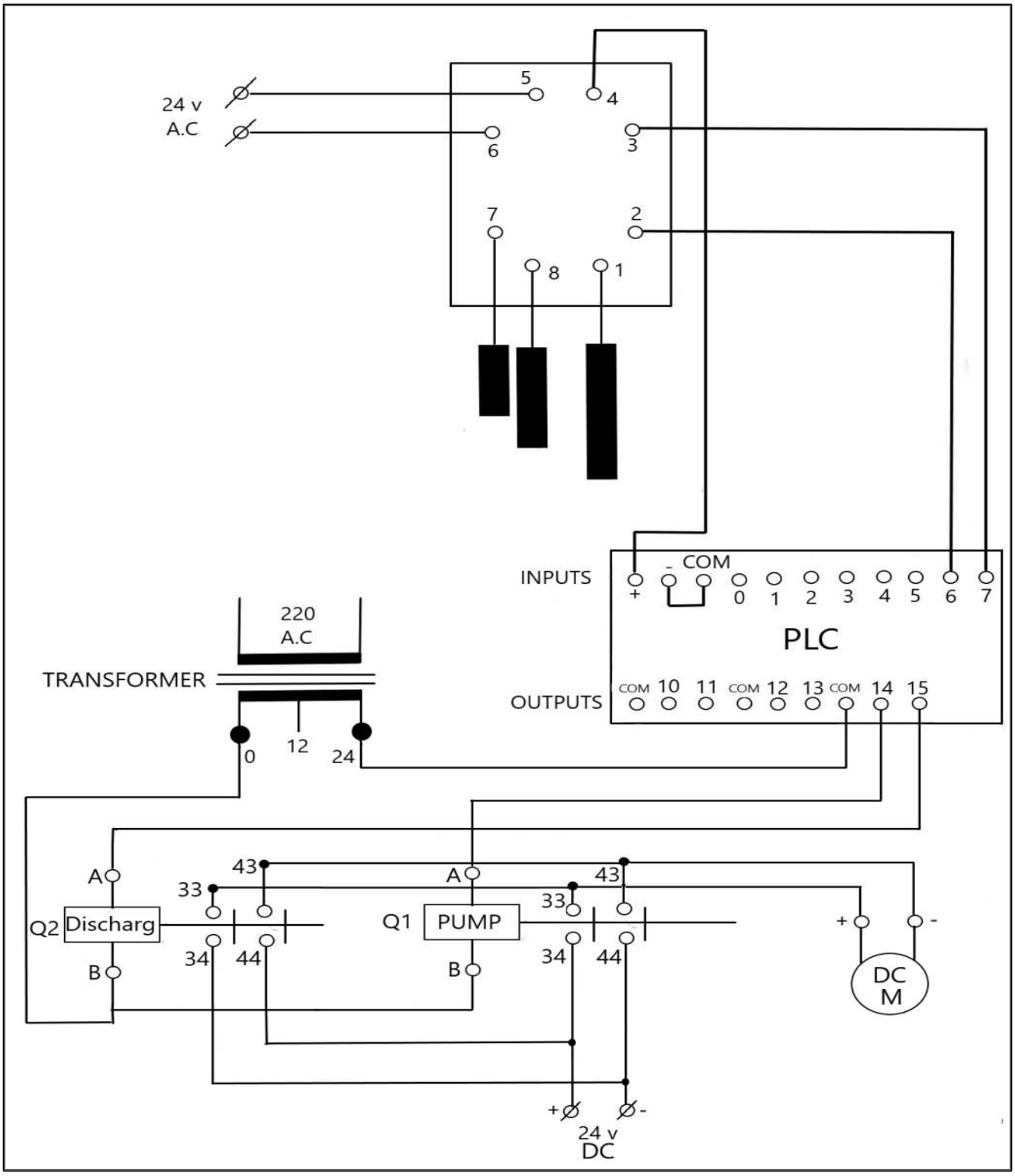

24 v
A.C
5
4
6
3
7
2
8
1
INPUTS
COM
+
0
1
2
3
4
5
6
7
PLC
220
A.C
TRANSFORMER
OUTPUTS
COM
10
11
COM
12
13
COM
14
15
0
12
24
A
43
33
Q2
Discharg
34
44
B
A
43
33
Q1
PUMP
34
44
B
+
-
DC
M
+
-
24 v
DC

Fig (1-3)

❖ Activity LAB:

The executive program of the control system shall be developed so that the following steps are accomplished respectively:

1. When we press the pushbutton and then after a certain period of time (five seconds) the water is transferred from the first tank to the second tank.
2. When the water reaches the high level the pump stops working for a certain period of five seconds.
3. After that, the pump starts to rotate in the opposite direction and starts to pump water from the second tank to the first tank.
4. When the water reaches the low level the pump will stop working for a certain period of time (5 seconds) and then start the process of pumping water from the first tank to the second tank as mentioned in the first item.

❖ Produce:

1. The electrical connections are tracked by the teacher to guide the student on the inputs and outputs.
2. Write ladder diagram to sequence program the previous.
3. Enter the program through “Manual Programmer ELC2000PL” as requested by the teacher.
4. Execute the program as per the following instructions.

❖ **HOME ASSIGNMENTS:**

Draw the block diagram and signal flow graph as in the Fig (1-4)

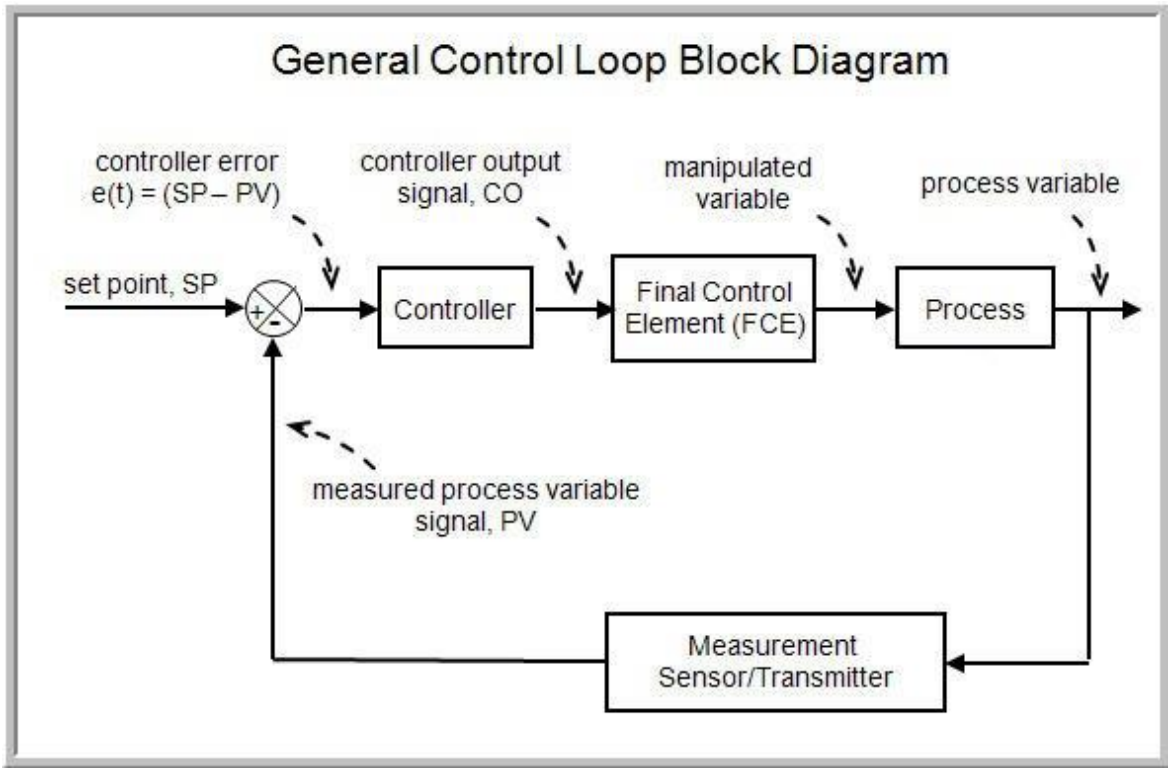


Fig (1-4)

Manual Programmer PLC ELC 2001 PL

- **Prepare your PLC for programming:**

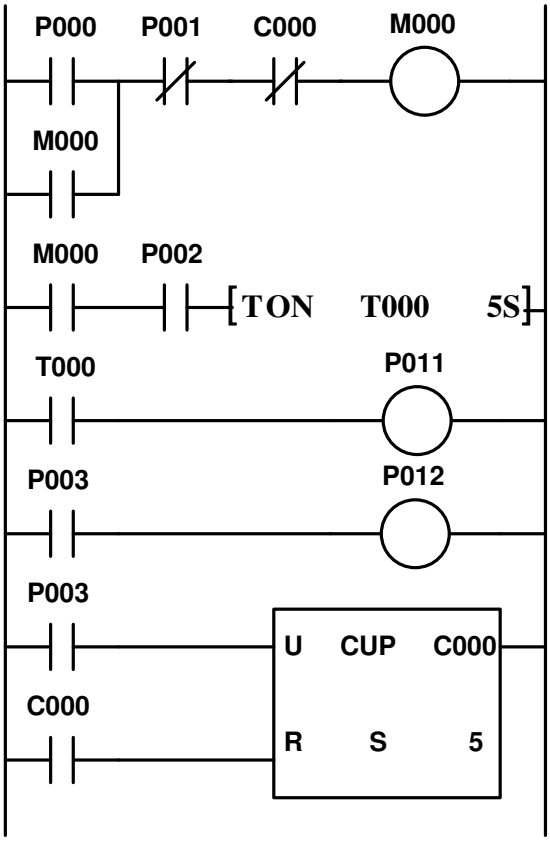
1. Run power supply.

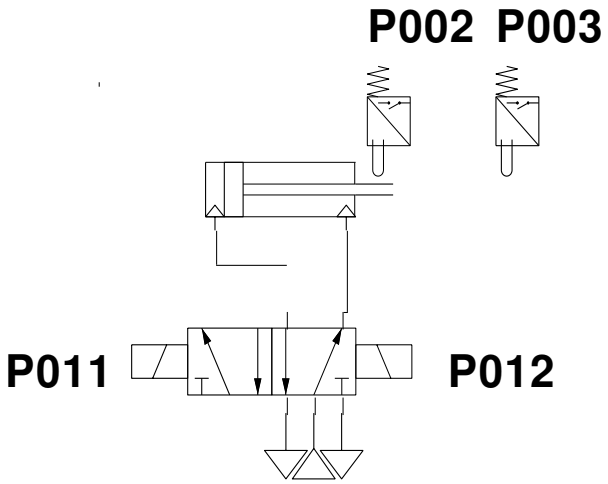
2. Before edition your program the programmer must be in program mode:
To do this press Key "1", to select "Programmer" and then press "ENTER".

3. You have to clear the PLC memory and the program memory:
To do this press key "SHFT + CLR" and then press key "SHFT + DELT".

4. Edit your program:

a. Enter your program, for example:





LOAD P000 (ENT)
OR M000 (ENT)
ANDNOT P001 (ENT)
AND NOT C000 (ENT)
OUT M000 (ENT)
Load M000 (ENT)
AND P002 (ENT)
TMR T000 (ENT)
DATA 00050 (ENT)
LOAD T000 (ENT)
OUT P011 (ENT)
LOAD P003 (ENT)
OUT P012 (ENT)
LOAD P003 (ENT)
LOAD C000 (ENT)
CNT C000 (ENT)
DATA 0005 (ENT)
END

b. To end your program:

Press key “FUN” and then press key “1” and then press key “ENTER”.

5. Load your program to the PLC:
To do this press key “RUN”







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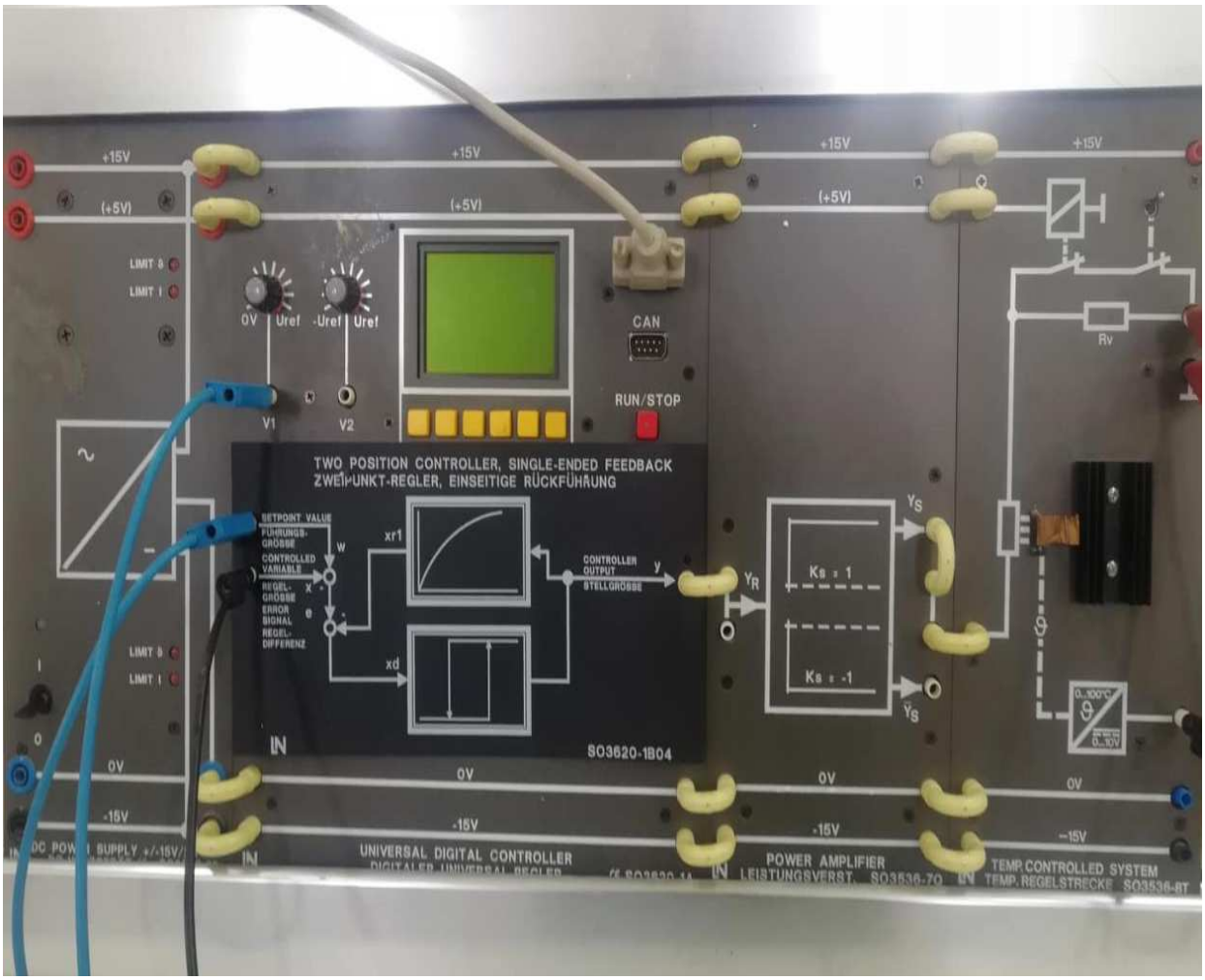
Ex.2

Two-position discrete controllers of the temperature (UNIVERSAL DIGITAL)

Student's Name:

Supervisor:

Date:


+15V
(+5V)
LIMIT ∂
LIMIT I
0V
Uref
-Uref
Uref
V1
V2
CAN
RUN/STOP
Rv
TWO POSITION CONTROLLER, SINGLE-ENDED FEEDBACK
ZWEIPUNKT-REGLER, EINSEITIGE RÜCKFÜHRUNG
SETPOINT VALUE
FÜHRUNGS-GRÖSSE
CONTROLLED VARIABLE
REGEL-GRÖSSE
ERROR SIGNAL
REGEL-DIFFERENZ
w
x
e
xr1
xd
CONTROLLER OUTPUT
STELLGRÖSSE
y
YR
Ks = 1
Ks = -1
Ys
SO3620-1B04
IN
-15V
UNIVERSAL DIGITAL CONTROLLER
POWER AMPLIFIER
LEISTUNGSVERST. SO3536-7C
TEMP.CONTROLLED SYSTEM
TEMP.REGELSTRECKE SO3536-8T

❖ Objective:

1. Identify the characteristics of (two position controller).
2. Identify the industrial applications in which this controller is used.
3. Compare this controller with other types of controllers.
4. How to organize a process variable through control variable.
5. Evaluation of the “step response” of the temperature process.
6. Find the transfer function of the temperature process.

❖ Introduction:

A two-position controller, also called an ON/OFF controller, is the most basic type of process controller. As the name implies, it provides an ON/OFF signal to the process’s control element see Fig (2-1). A typical example of an ON/OFF controller is a home heating system. The heater turns ON when the temperature is below the set point and turns OFF when the temperature reaches an acceptable level. Ideally, if the set point temperature is 70°F, the heater will turn ON when the temperature is less than 70°F and turn OFF when it is greater than 70°F, as the heater tries to keep the error (SP – PV) at zero.

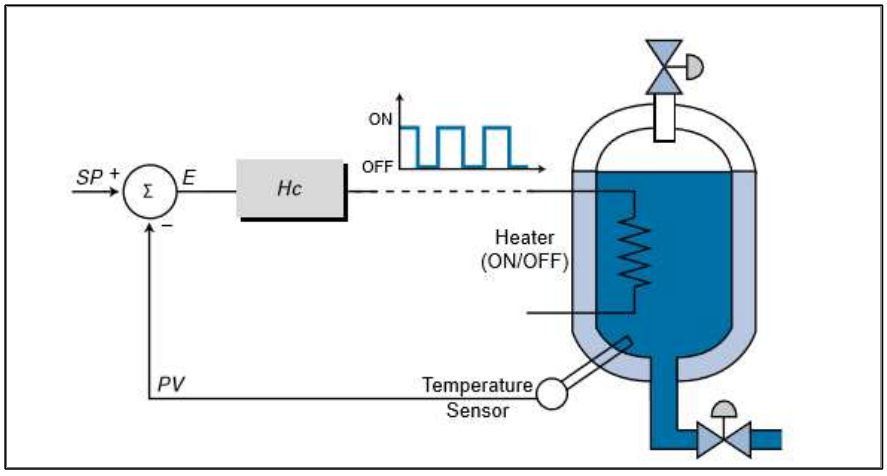


Fig (2-1)

However, most heating systems have an error dead band, meaning that the heater will turn OFF at a value just above the target temperature and turn ON at a value just below it. So, if the heater in our example has a dead band of 68°F to 72°F, the heater will turn OFF when the temperature reaches 72°F and turn ON when it falls to 68°F. This dead band range avoids the constant ON/OFF action associated with trying to keep the process variable at one exact set point.

The example heating system has a reverse-acting controller, because if the controller senses that the process variable (temperature) decreases, it will increase its output to 100% (ON), as shown in (Figure 2-2). As a result of the error dead band, the heater will turn ON when the temperature drops to 68°F and turn OFF when it reaches 72°F. Furthermore, note that the controller is sometimes ON and sometimes OFF within the error dead band. This depends on the direction of the process variable. If the value of the process variable is decreasing within the dead band, then the controller is OFF, since it senses that the temperature is still at an acceptable level (reverse-acting). When the process variable reaches the lower limit of the dead band, then the controller will turn ON, causing the direction of the process variable to change (i.e., to increase). The controller will remain ON until PV reaches the upper limit of the dead band. At that time, the controller will turn OFF, PV will begin to decrease, and the cycle will repeat.

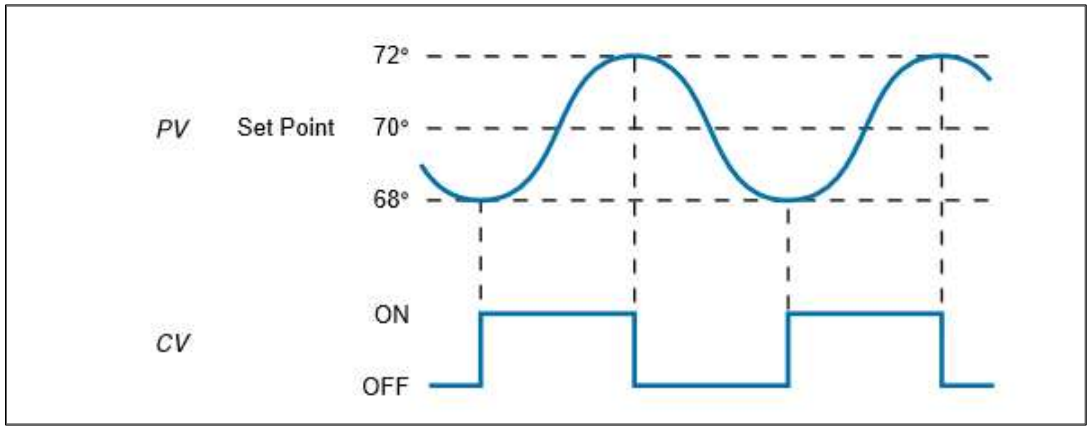


Fig (2-2)

The action of an ON/OFF controller can be described by:

$$CV = 100\%\,(\text{ON}) \quad \text{IF error} > -\Delta E$$
$$CV = 0\%\,(\text{OFF}) \quad \text{IF error} < +\Delta E$$

Where: ±ΔE represents the error dead band. Graphically, this controller output can be represented as shown in (Figure 2-3), where the dead band of ±ΔE is equal to ±2°F. If the process variable "comes" from the positive side (i.e.) the error declines to a value less than –ΔE, or 68°F), the controller output will turn ON at (point 1) and remain ON (point 2) until the error reaches +ΔE (point 3).

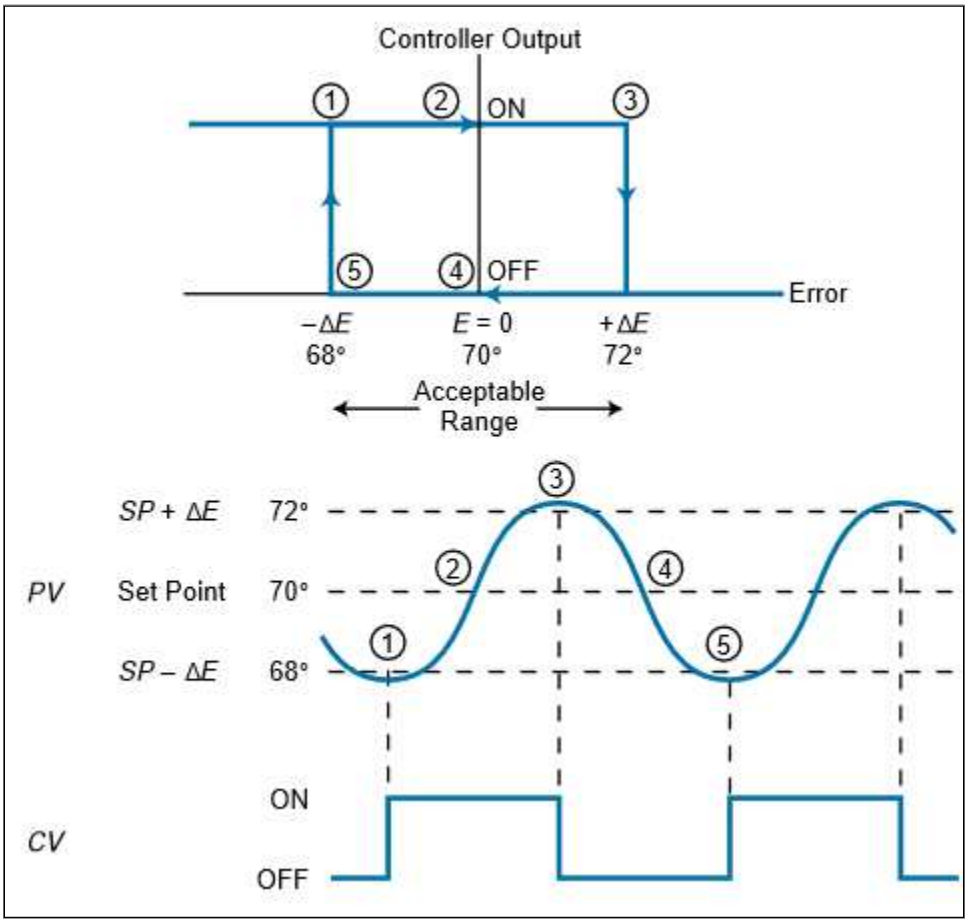


Fig (2-3)

At that time, the controller will turn OFF and remain OFF (through point 4) until the error drops to –ΔE (point 5), causing the cycle to repeat. This dead band curve is said to have hysteresis, meaning that the reaction of the system depends on its previous actions.

It also produces an oscillating response, which is acceptable in this case. Also note that the curve of the ON/OFF controller signal will tend to overshoot the SP + ΔE value and undershoot the SP – ΔE value of the heater system due to finite warm up and cool off times (lag times).

ON/OFF controllers are appropriate for applications where large-scale, sudden changes are uncommon and the process reaction rate is slow. If the error dead band of the controller is reduced, then the amount of error in the system will decrease; however, the frequency of the ON/OFF and process variable cycles will increase. Conversely, if the dead band is increased, the oscillation frequency will decrease, but the error will be maximized. Thus, a trade-off exists between the desired error dead band and the frequency of the ON/OFF activation of the control element. The control element (e.g., valve, compressor, etc.) and other system components may be seriously damaged if they are turned ON and OFF too rapidly. Therefore, the system must be configured to compromise between the error allowance and the frequency of oscillation

❖ Necessary Equipment:

1. Computer.
2. Universal digital controller (Consists of):
   - Supply voltage , +15 volt:
     Maximum current consumption here is 600 mA.
   - Connection rail for 5 v supply voltage:
     This rail is used for through connection of a 5 v supply that may be available. The 5 v is not required for this controller panel.

- Supply voltage, ground rail:
  Socket for power supply: earth (0v) connection.
- Supply voltage,-15volt:
  Socket for connecting the negative supply voltage
  Maximum current consumption here is 60 mA.
- Voltage output (v1,v2):
  2 potentiometers for adjusting the programmable reference voltages, that are available at 4mm safety socket.
- LC-display:
  With graphic capabilities, LC-display with background illumination resolution 128*64 dots
- Serial interface RS232:
  This socket must be connected to the serial interface when operating with a PC. Use a standard 9-pole cable.
- Pushbutton for switching the controller ON or OFF (RUN/STOP):
  Operation of the controller is indicated by a LED integrated in the pushbutton.
- Operating keypad:
  Six keys for adjustments and operating the controller
- Analog inputs(W,X):
  Four analog inputs with measurement range of ±10v, ±5v or ±2-5v, adjustable by software Optional auto ranging facility 10 bit resolution. Input impedance 10 KΩ Maximum permissible input voltage (± 15v).
- Analog outputs (y):
  4mm safety sockets. Maximum output voltage, ±10v. Status indication via LED (overdrive) .maximum output current, 4˜10 mA.

❖ Procedure:

1. Turn on the computer.
2. open program "tow position controller" from desktop
3. Press this icon  from the toolbar.
4. From “sittings” Press “controller parameter”, from the list "control input" press "manual".
5. From “chart” press “properties”.
   A. from the list “signals” Select the following signals:
      - “Control input” Choose a specific color.
      - “Controlled variable” Choose a specific color.
      - “Upper switching point” &“lower switching point” Choose a specific color.
   B. from the list “scale” Select the following:
      - “Maximum” writ “200”.
      - “Division” writ “10”.
6. Apply the following table.

hysteresis	Set point		
	25	35	50
	5	5	5
	10	10	10
	20	20	20
	50	50	50

❖ **HOME ASSIGNMENTS:**

Draw the block diagram and signal flow graph as in the Fig (2-4)

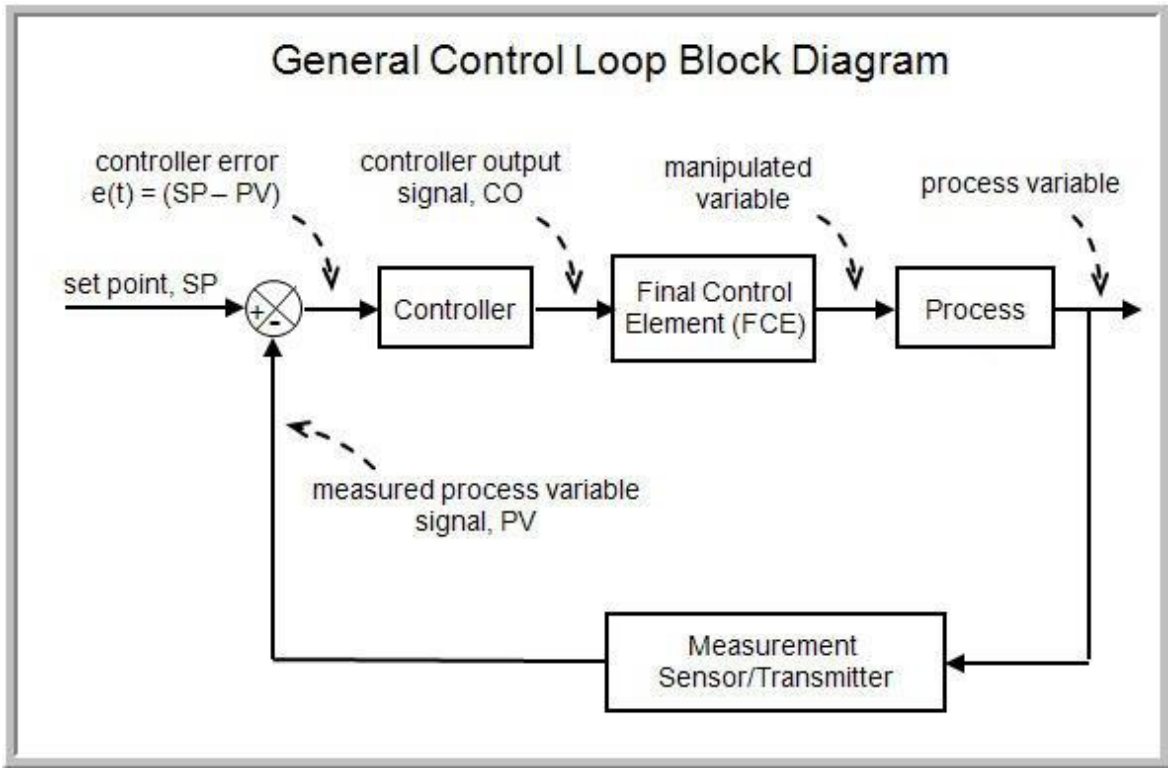


Fig (2-4)

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Ex.3

PID controllers of the Pressure

(UCP-P)

Student's Name:

Supervisor:

Date:

❖ Objective:

1. Monitoring the optimization process of a PID controller for a given process.
2. Become familiar with the most commonly used methods for optimizing the variables of a PID controller from the characterization of the process.

❖ Necessary Equipment:

1. Control valve.
2. Pneumatic valve.
3. Relative pressure manometer.
4. Absolute pressure sensor.
5. Storage tank.
6. Inlet/outlet valve.
7. Diaphragm.
8. Exhaust valve.
9. Pointer flow measurer.
10. Differential pressure measurer.
11. I/P converter.
12. Control interface.

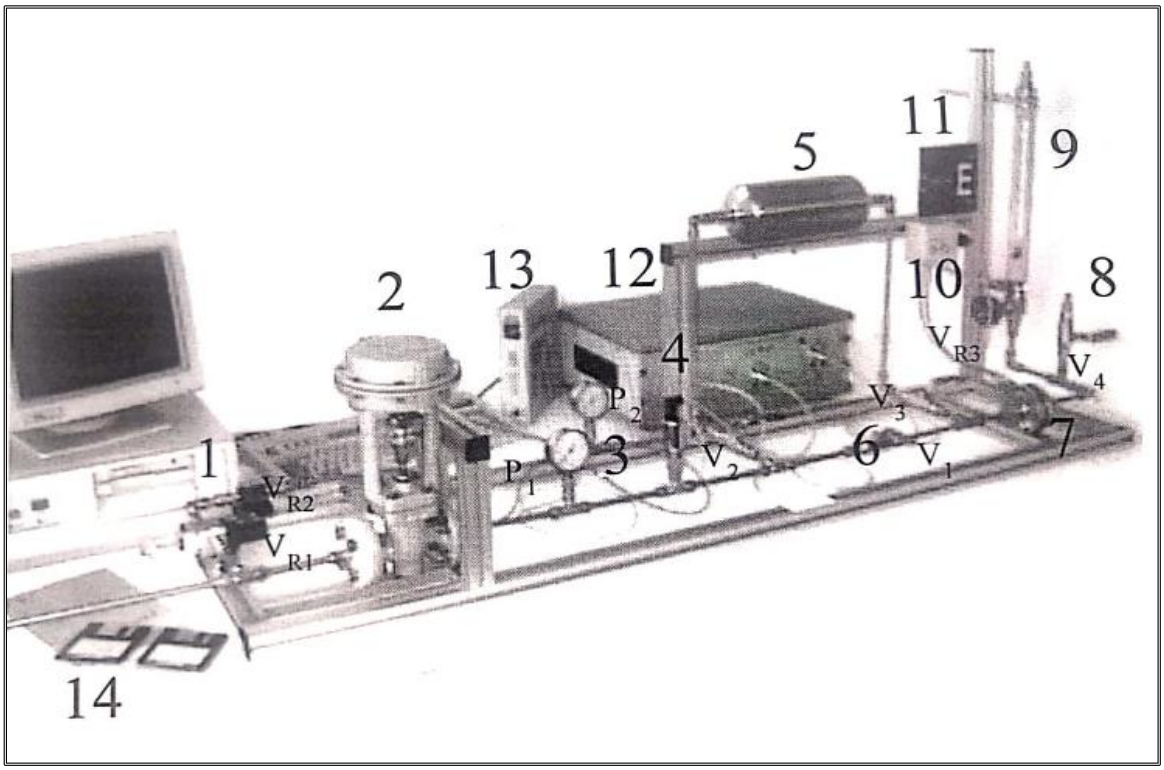


Fig (3-1)

❖ **Description of the unit:**

The unit includes an A/D analogical-digital converter card with 16 inlet channels, which permit the inlet of signals emitted by 16 different sensors. The card also has two analogical outlet channels, 0-5V or 0-10V depending on the position of the internal jumpers which will be used to control the pneumatic valve with the cur- rent-pressure converter (I/P).

A fat wire connects the A/D converter card to the interface, which adapts the signals emitted or received by the computer, the sensors or the actuator. The signal from the absolute pressure, the differential pressure, and I/P converter go the interface. The pressure signals (either absolute or differential) are current signals between 4 and 20 mA and are turned by the interface into tension between 0 and 10 V .A current signal is less sensitive to noise than a voltage one, especially if the latter is a low voltage one. What the industry normally does is to transfer the signal in current through to long enough wire and then turn it into voltage for their use in control.

For the correct work of the unit an air source or a compressed gas source will be required, being the compressed gas source completely clean, without water, oil or particles. This source must be able to create a flow rate 4 m^3 /h with a maximum pressure of 2.5 bar. Nevertheless, it is advisable to use the unit with lower figures.

If the compressed air source required is not able to create the necessary flow rate, the ball valve V4 shall be closed and the pressure inside the system shall be manually controlled with the pressure gauge VR3.

❖ **Description of the element:**

1. Adjusting valve:



These valves are placed at the entrance and at the exit of the primary and secondary circuit. Its function is to control the pressures at the entrance of both pipes. The valve located in the secondary system also has a filter which blocks the way to dirt into the I/P converter. These valves let the flow of air pass through and

therefore, let the pressure raise when the valves turn in a clockwise direction. On the other hand, when turning an anticlockwise direction, the pressure and the flow of air are stopped.

2. I/Converters:

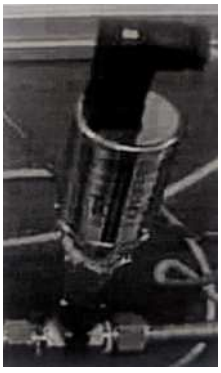
   It is a blocking mechanism, in which the closing and opening systems are controlled by a membrane linked to a small magnet inside a solenoid. When an electric current flows inside the solenoid a magnetic field is created. The magnetic field is proportional to the current intensity which circulates inside the coil. When an electric current circulates through the solenoid a magnetic field is created, which is proportional to the intensity of the current which circulates through the coil. This produces a movement of the permanent magnet which moves the membrane and both create an exit pressure proportional to the supplying current. With a 4-20 ma entrance current a 0.2-1 bar linear exit pressure is created with a 1% precision.

3. Pneumatic valve:

   This valve opens and closes the inlet of the liquid which flows along the lower area by controlling the pressure at the entrance of the top area with an I/P converter which at the same time controls the diaphragm. For easiness reasons, it is air the control- ling fluid and also the fluid that flows inside the valve, although they could be different and even the valve would control the pass of liquids. A valve must always work under the most severe safety conditions, and even a cut of the flow of air or of the electric supply shall be foreseen.

4. Piezoelectric pressure sensor:

   Piezoelectric elements with which units are assembled for measuring absolute pressures are crystalline materials which, when deforming as a consequence of pressure, emit an electric signal. Quartz and barium titanate are materials typically used for this kind of sensors as they are able to endure temperatures of 150°C. They are light, compact, robust and small. The exit signal is linear and proportional to the are supply. They are able to measure dynamic



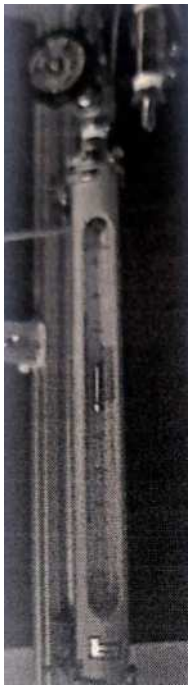
pressures under a frequency of 10° Hz. and its answer is immediate. With an electric supply of 24V CC the linear tension signal emitted is between 0 and 5 V for a pressure from 0 to 2.5 at.

5. Differential pressure sensor:



The differential pressure sensor has two cameras for high and low tension, which are separated a membrane made with a piezoresitant material. The pressure difference in both cameras causes a fault in the Westinghouse bridge which produces a voltage signal. The differential pressure sensor supplied with the unit has a size from 2.5 to 69 mbar and an exit between -2.5 and 16 mV, with a 0.3% precision and it can work with temperatures of even 85°C.

6. Flowmeter A flowmeter:



Is a widely used instrument in pipes for air for measuring the flow rate of air. This instrument includes a ruler divided into millimeters from 0 to 100% and a flowmeter of a specific weight. Its weight establishes the flow rate in a pipe depending on its position. Therefore, the measure of the scale depends on the weight. The flowmeter supplied with the unit has a maximum flow rate of 9.5 m^3 /h and a minimum flow of 0.8 m^3/h. This means that to read the position of the de 15% flowmeter there is not a volume of 1.45 m^3 /h (24.16 1/min).

❖ CONTINUOUS-MODE CONTROLLERS:

Most process control applications employ continuous-mode controllers, instead of discrete-mode controllers, to avoid the oscillatory system response caused by ON/OFF control. A continuous-mode controller sends an analog signal to the process control field device (see Figure 3-2) to regulate the process variable, bringing the error signal to zero in a closed-loop system. A continuous-mode controller behaves like a multiposition controller with an infinite number of positions.

Continuous-mode controllers use three different modes to control the process:

- Proportional control mode
- Integral control mode
- Derivative control mode

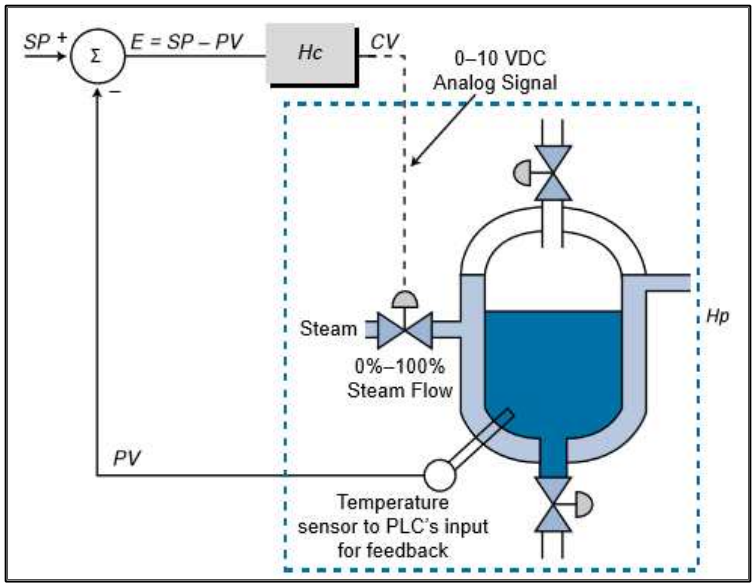


Fig (3-2)

These three modes are also referred to as controller actions, each one reacting differently to the error present in the system in a direct- or reverse-acting fashion. The proportional mode provides a control variable adjustment that is proportional to the error deviation. The integral mode (or reset mode) provides a change in the control variable based on the time history of the error. The derivative mode (or rate mode) provides a change in the control variable based on the rate of change of the error signal. The combination of all three modes in one controller forms the industry standard known as PID control. Table 3-1 shows the different possible combinations of continuous controller

modes. Note that the derivative action is not used as a stand-alone mode in applications. This is due to the derivative action's response, which produces a high output but only for a short period of time. This has little effect on the process and, consequently, does not provide any process control.

Controller Mode		Response	Applications
Proportional	P	*CV* changes in proportion to *E*	Systems with small load changes and/or small to moderate lag times
Integral	I	*CV* changes according to how *E* changes over time	Processes with small process lags and small capacities
Derivative	D	*CV* changes according to how fast *E* changes	Not used alone in applications
Proportional-Integral	PI	*CV* responds in a combination of P and I actions	Systems with large load changes
Proportional-Derivative	PD	*CV* responds in a combination of P and D actions	Processes with fast load changes
Proportional-Integral-Derivative	PID	*CV* responds in a combination of P, I, and D actions	Can be used in practically all process control applications

Table (3-1)

A. PROPORTIONAL CONTROLLERS (P MODE):

A proportional controller adjusts the control variable output in a manner proportional to the error. As shown in Figure (3-1), the controller (Hc) receives feedback information from the process (Hp) in the form of the process variable, which is then compared to the set point. The error created, either positive or negative, tells the controller what percentage of output (CV) to provide to bring the error to zero. Figure (3-3) illustrates a typical proportional controller transfer function for a direct-acting controller (e.g., a cooling system). As the error becomes more negative (PV > SP), the controller will increase the control variable in proportion to the error.

This will cause the process variable (from Hp) to decrease, thus pushing the error to zero. If the error becomes more positive, the opposite occurs.

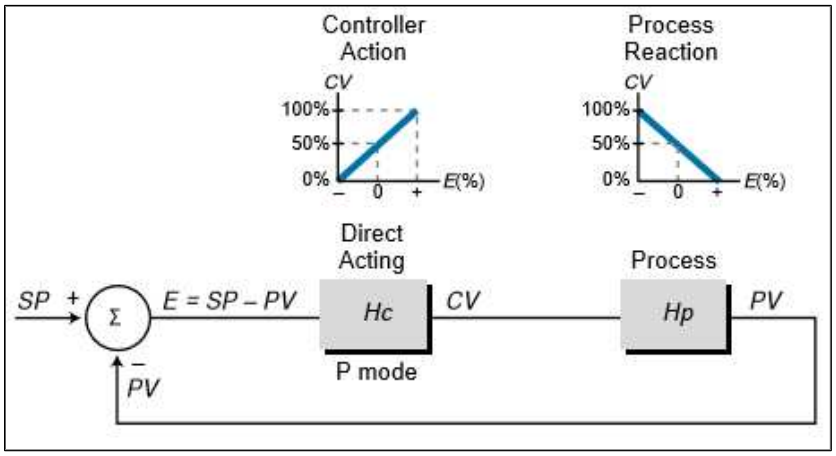


Fig (3-3)

The control variable output (CV (t)) of a proportional controller, starting from the set point value, is expressed by:

Where: $CV_{(t)} = K_p E + CV_{(E=0)}$

K_P = the proportional gain of the controller.

E=the current error

$CV_{(E=0)}$ = the controller output when the error equals 0

Using this equation, a proportional controller can adjust the value of the control variable according to time and error by replacing the CV (E=0) term with the previous value of CV:

$CV_{new} = K_P E + CV_{old}$

So, if a controller with a control variable output value of 50% senses that the proportional error in the system (KPE) is 20%, its new output will be 70%:

$$CV_{new} = 20\% + 50\%$$

$$= 70\%$$

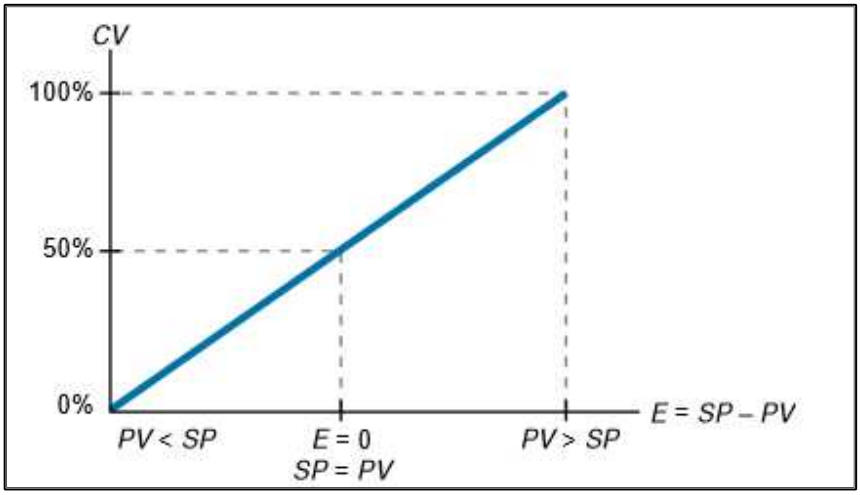


Fig (3-4)

This value indicates a linear correspondence between the control variable and the error, as was depicted in the graph in Figure (3-4). This graphic representation is called the proportional band, and it shows the error values associated with the full range of the controller output. The slope of this graph, the proportional gain KP, is computed by dividing the percentage change in output by the percentage change in error:

$$KP = \frac{\%\ change\ in\ CV}{\%\ change\ in\ E}$$

The proportional gain, therefore, is expressed in units of %/%. For example, a gain of 1 indicates that a 1% change in error will cause a 1% change in controller output. Note that the direction of the slope of the proportional gain (the positive or negative response of the control variable to a change in the error) depends on whether the controller is direct acting or reverse acting.

The proportional gain relationship between the error and the control variable depends on the width of the band upon which the controller is acting.

B. INTEGRAL CONTROLLERS (I MODE):

An integral controller provides an output whose rate of change is proportional to the error deviation. This means that the larger the error, the faster the controller's output changes and vice versa. An integral controller will stop adjusting its output once the error becomes zero. When used in conjunction with a proportional controller, an integral controller will bring the system's residual error to zero. An integral controller's output (CV) is represented by:

$$\frac{dCV}{dt} = K_I E.$$

Where:

$\frac{dCV}{dt}$ = the rate of change in controller output in % over seconds.

K_I= the integral gain in % of the controller output per second per % error.

E = the error in %.

This differential equation indicates that the controller's output CV(t) can be obtained by taking the integral over time of both sides of the equation so that:

$$\frac{dCV}{dt} = K_I E$$

$$dCV = K_I E dt$$

$$\int dCV = \int K I E dt$$

$$CV_t - CV_{(t=0)} = K_I \int E dt$$

$$CV_t = K_I \int E dt + CV_{(t=0)}$$

Where the term $CV_{(t=0)}$ is the value of the output at t = 0. When an integral controller is used in a closed-loop system, it calculates $CV_{(t)}$ for every change in error. So, if the value of the error changes after the controller has calculated a previous value $CV_{(t)}$, then

it will use this previous value of $CV_{(t)}$ as the $CV_{(t=0)}$ value and calculate a new $CV_{(t)}$ output based on the new error.

The integral gain K_I (see Figure 3-5) indicates the sensitivity of the output's rate of change to the percentage of error that occurs over time. A large value of K_I means that a small error will produce a large rate of change in the controller output. Conversely, a small value of K_I means that a small error produces a small rate of change in the controller output. In Figure (3-5), the rate of change of K_{I1} is greater than that of K_{I2}.

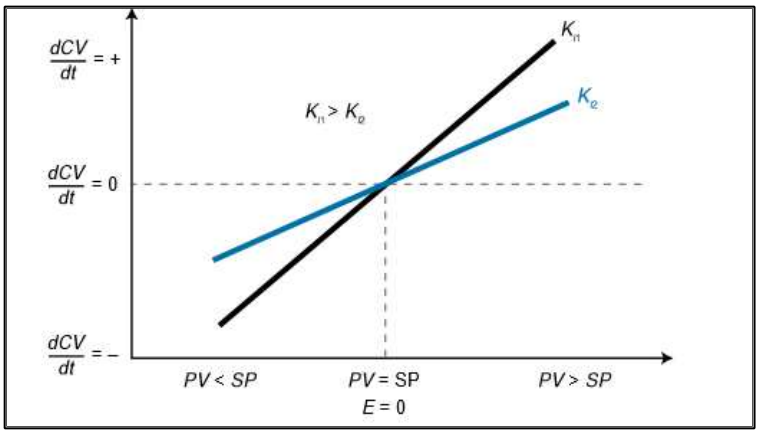


Fig (3-5)

The gain of an integral controller (KI) is defined by the equation:

$$K_I = \frac{(\frac{dCV}{dt})}{E}$$

$$= \frac{change\ in\ \%\ of\ per\ second}{change\ in\ \%\ of\ error}$$

A value of KI = 0.2 indicates that the controller will change 0.2% per second for every 1% of error present in the system. So, if the 1% error in the system lasts for 2 seconds and then goes to zero, the controller will increase its output 0.4%.

The inverse of the gain term KI is referred to as the integral time (TI), or reset time, in seconds. The integral time is the time it takes for the control variable (CV) to change 1% for a 1% change in error. It is expressed as:

$$T_I = \frac{1}{KI}$$

C. PROPORTIONAL-INTEGRAL CONTROLLERS (PI MODE):

Although an integral controller does not have the residual error at steady state that a proportional controller has, its response action to a step change in input (step in error) is often too slow to be used in real-life applications. This slow speed, as compared with the immediate response of a proportional controller, is due to the ramping effect of the integral action as the controller increases its output. Therefore, proportional action is normally added to an integral controller (see Figure 3-6) to form a proportional-integral (PI) controller. This type of controller has a fast response time (proportional action), plus it eliminates all residual error (integral action). As illustrated in Figure (3-7), a PI controller can have one of two configurations:

- Parallel
- Series

In a parallel PI controller, the proportional and integral actions occur independently of each other, so the controller's output (CV) is equal to the proportional action plus the integral action:

$$CV_{new} = K_P E + K_I \int Edt + CV_{old}$$

In a series PI controller, on the other hand, the integral action occurs after the proportional action. Therefore, the input to the integral action is not the system error E, but rather the result of the proportional action KPE. Accordingly, a series PI controller's output is defined by:

$$\begin{aligned} CV_{new} &= K_P E + K_I \int KpEdt + CV_{old} \\ &= K_P E + K_P K_I \int Edt + CV_{old} \end{aligned}$$

Both of these types of P_I controllers eliminate error offset and have a faster response time than an integral-only controller. However, series P_I controllers multiply the integral gain times the proportional gain, producing an effect called repeating. In repeating, the effect of the proportional gain (K_{PE}) is repeated during every integral time period T_I, causing the integral action of the controller to equal that of the proportional action. This means that a series PI controller responds faster to a change in error than a parallel

controller when their proportional gains are greater than one ($K_P > 1$) and their integral times are the same.

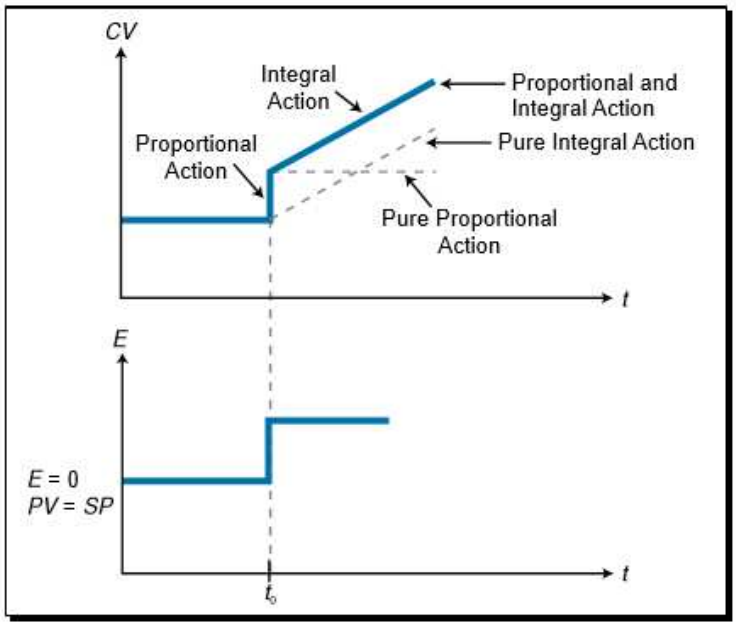


Fig (3-6)

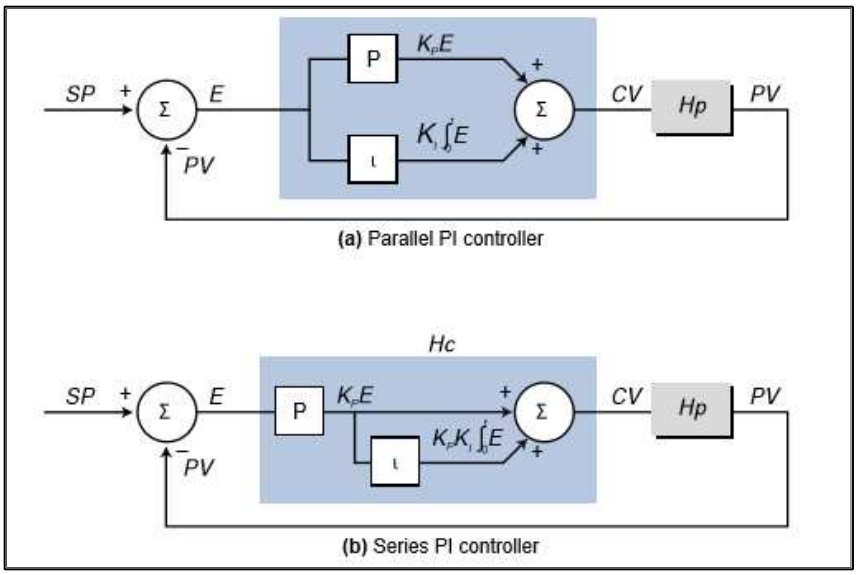


Fig (3-7)

The term repeats is used when referring to how many times the proportional amount is repeated in one minute. If the value of T_I is less than 1 minute, then the integral gain is repeated more than one time per minute.

When t = T_I, the term KPE is repeated once by the integral action, in this case in the period from t = 0 to t = 1 minute. Figure (3-8- a) illustrates the integral gain of a P_I controller with a repeat of 1 (i.e., $T_I = 1$). After 1 minute, the term K_{PE} is repeated. Figure (3-8- B) illustrates a P_I controller with an integral time of $T_I = 0.333$, indicating that the term K_{PE} will be repeated three times in one minute.

D. DERIVATIVE CONTROLLERS (D MODE):

The output of a derivative controller is proportional to the rate of change of the error in the system, which is expressed as $\frac{dE}{dt}$ (see Figure 3-9). This derivative action, also referred to as rate mode, is expressed mathematically as:

$$CV_{new} = K_D \frac{dE}{dt} + CV_{old}$$

Where:

CV_{new} = the control variable.

CV_{old} = the previous value of CV.

K_D = the derivative gain constant in %(sec/%)

$\frac{dE}{dt}$ = The rate of change of error over the duration of change in %/sec

The derivative gain constant (K_D) is also referred to as the rate time. It can be expressed in seconds or minutes as:

$K_D = T_D$ seconds (rate time)

Or

$K_D = \frac{Td}{60}$ minutes (if T_D is given in seconds)

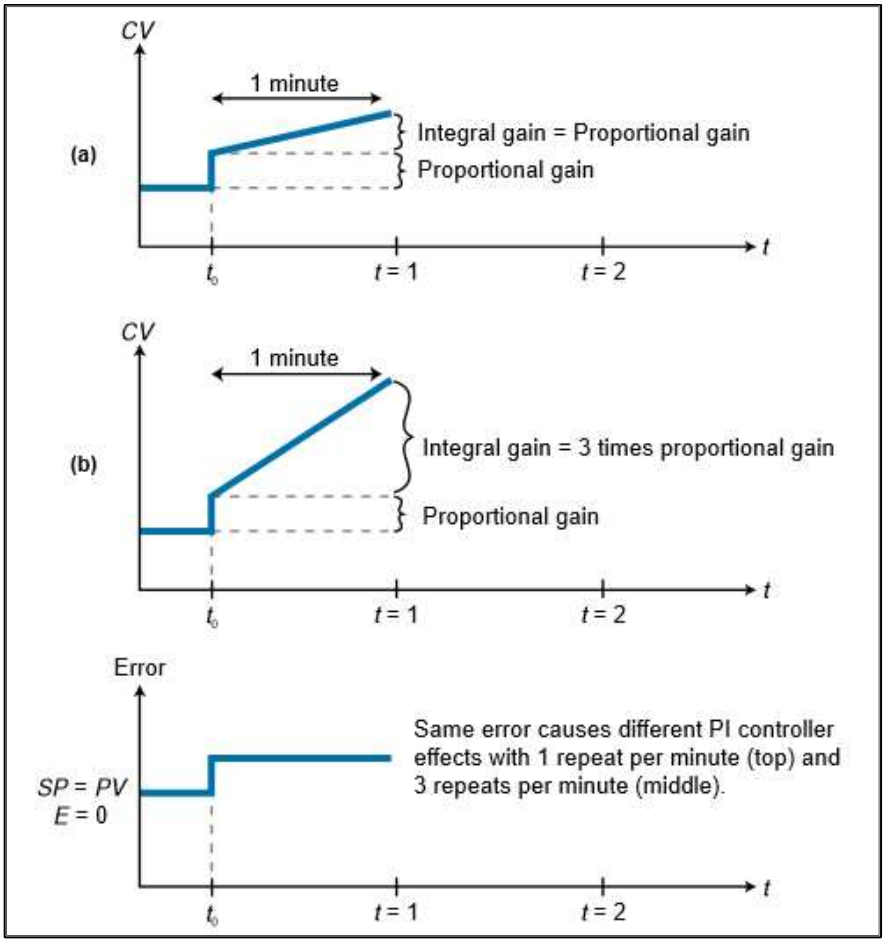


Fig (3-8)

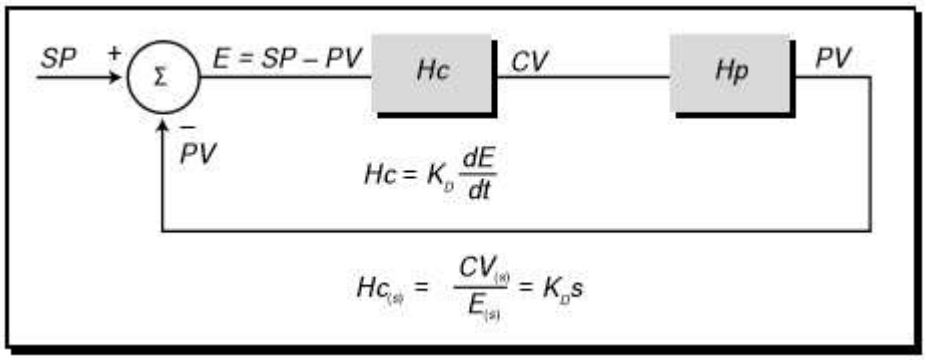


Fig (3-9)

In Laplace form, the derivative controller transfer function takes the form:

$$Hc_{(s)} = \frac{CV\ (s)}{E\ (s)}$$
$$= K_{DS} = T_{DS}$$

Figure (3-10) illustrates the derivative gain transfer function in a direct acting system by indicating the corresponding controller outputs for different rates of change ($\frac{DE}{dt}$) in error. Like in the integral mode, the rates of error change form several family curves (see Figure 3-10 b). For example, if the error increases at a rate of 1.0%/sec, the controller will apply a derivative action that makes its output jump from 50% to 70% (see Figure 3-10 a). If the rate of increase slows down to 0.5%/sec, the controller will decrease its output to 60%. When the rate of change of error equals zero, the controller will decrease its output to 50% again (see Figure 3-11). Note that the derivative action is based on the rate at which the error changes, not the actual value of the error.

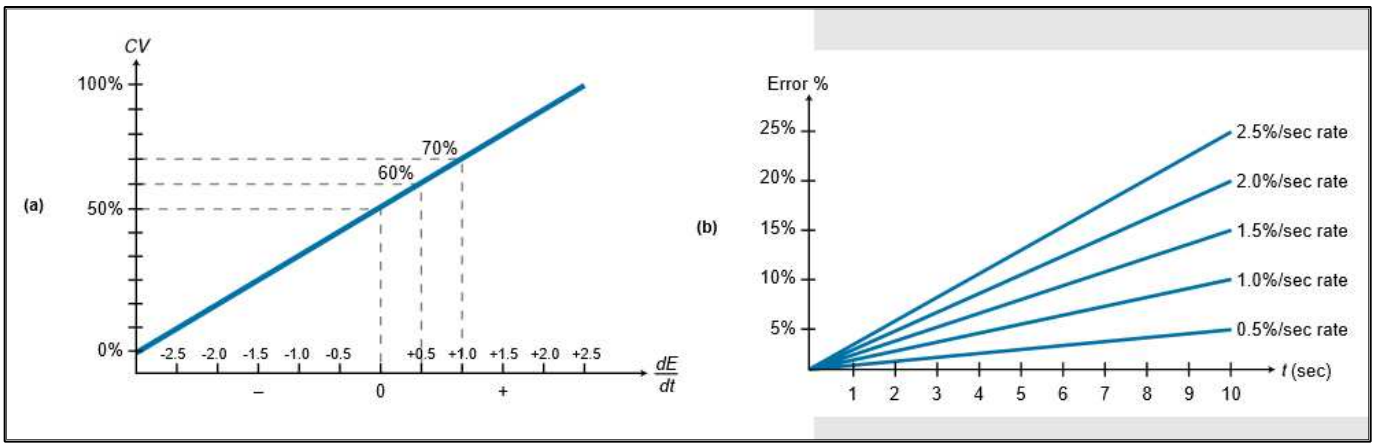


Fig (3-10)

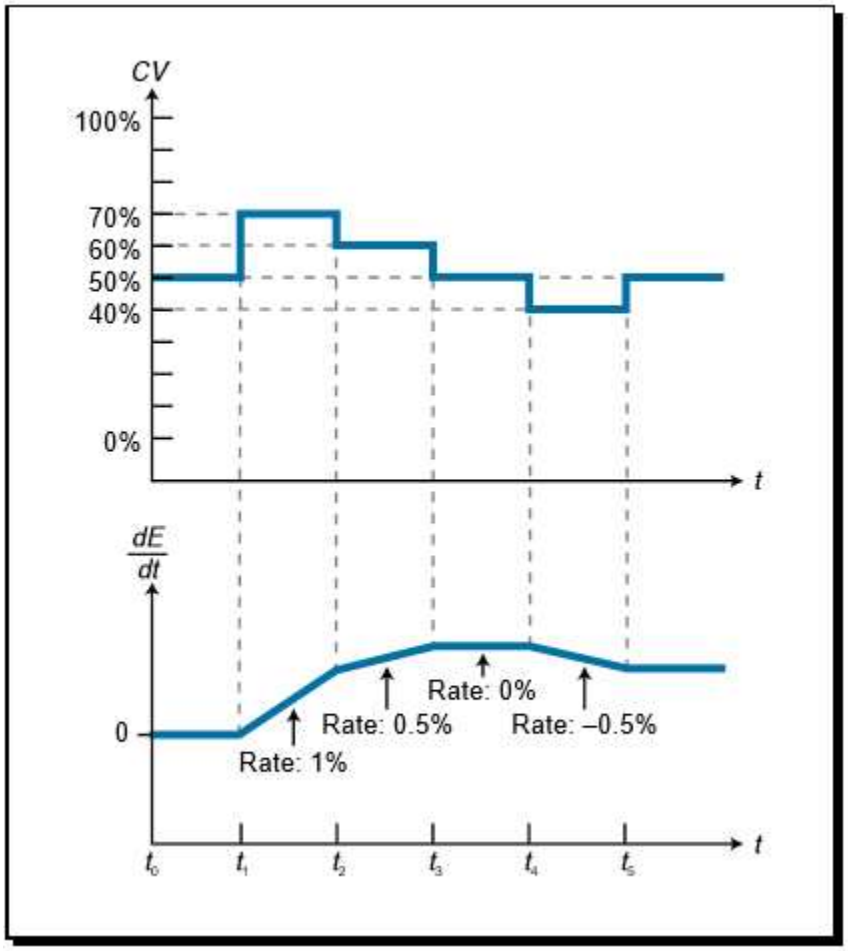


Fig (3-11)

E. PROPORTIONAL-DERIVATIVE CONTROLLERS (PD MODE):

The derivative action in a PD controller adds stability to a closed-loop system by reducing the amount of overshoot and undershoot in the system's response. The derivative component acts as a "brake" in the system, slowing the proportional response as the process variable approaches its set point. The speed of response, however, also slows down. To observe this braking effect, let's examine the reaction of the closed-loop system in Figure (3-12) to a unit step. This is a second-order system with a proportional gain of KP = 8 and no derivative gain (switch open). The addition of derivative action to this system will help to stabilize the overshoot and undershoot of the response to a change in error.

If the set point in Figure (3-12) changes, the proportional controller will try to bring the error to zero by making PV equal SP. The error at the start (t0) is 1, and as PV approaches SP, this error becomes smaller. In this proportional action, the controller output is positive (direct acting), which makes the PV value become more positive. The slope of PV is also positive, as seen at point A. This positive value of $\frac{\mathrm{dPV}}{\mathrm{dt}}$ can be approximated as shown in Figure (3-12). If derivative action were present in this system (switch closed), then the value of $\frac{\mathrm{dPV}}{\mathrm{dt}}$ would be negative:

$$\mathrm{PV}_{(t)} = \mathrm{K_P E} - \mathrm{K_P K_D}\,\frac{dPV}{dt} + \mathrm{PV}_{(t=0)}$$

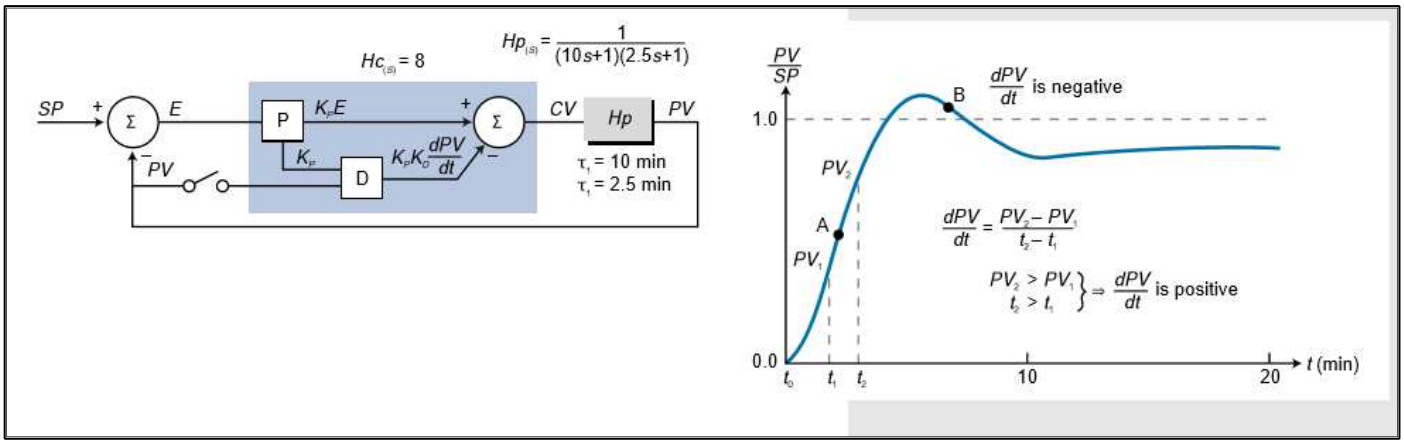


Fig (3-12)

Thus having the opposite sign of the proportional gain. Therefore, with derivative action, the output of PV (t) (at point A) would be less than a pure proportional controller without derivative action. In fact, a positive $\frac{\mathrm{dPV}}{\mathrm{dt}}$ term (slope) would make the derivative term in the PD system equation negative. This indicates that the derivative action of the PD controller will break the response of the proportional action, therefore reducing the amount of overshoot. The same holds true when the slope is negative, which occurs when the response of the pure proportional action starts to decrease (point B). When the proportional response becomes negative, the derivative term becomes positive, thus braking undershoot:

$$CV_{(t)} = K_P(-E) - K_P K_D\left(-\frac{dPV}{dt}\right) + CV_{(t@b)}$$

$$= -K_P E + K_P K_D\left(\frac{dPV}{dt}\right) + CV_{(t@b)}$$

So, by adding the derivative action to this closed-loop system (switch closed in block diagram), it is possible to reduce the overshoot and undershoot through the braking effect of the derivative action.

F. PROPORTIONAL-INTEGRAL-DERIVATIVE CONTROLLERS (PID MODE):

The basic expression for the process variable output for a standard parallel PID controller (see Figure 3-13) is:

$$PV_{(t)} = K_P E + K_I \int Edt + Kd\ \frac{dE}{dt} + PV_{(t=0)}$$

In Laplace form, this controller's transfer function is represented as:

$$Hc_{(s)} = \frac{CV(s)}{E(s)}$$

$$= K_P + \frac{Ki}{S} K_D\ s$$

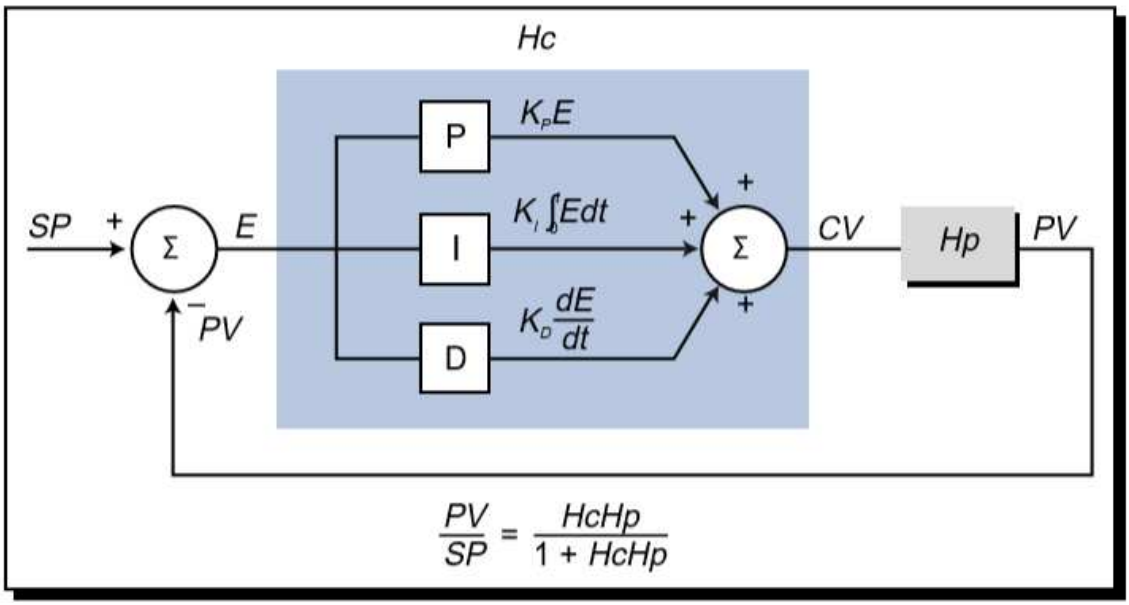


Fig (3-13)

❖ **Produce:**

1. Valves V_2, V_3 must be open and V_{R-3} must be 30% regulated. Valves V_1 and V_4 must be closed.
2. Open Pc and system software.
3. Adjust the "consign value "on ".75 bar".
4. Increase the proportional band from 10% to 20 % to 30 % to 50% until the system stable (P range 01% ~ 100%)
5. Add the integral range until the offset become zero (I range .1 ~ 10)
6. Add the differential range until the system is accelerated (D range .01 ~ .5).

❖ **HOME ASSIGNMENTS:**

Draw the block diagram and signal flow graph as in the Fig (3-14)

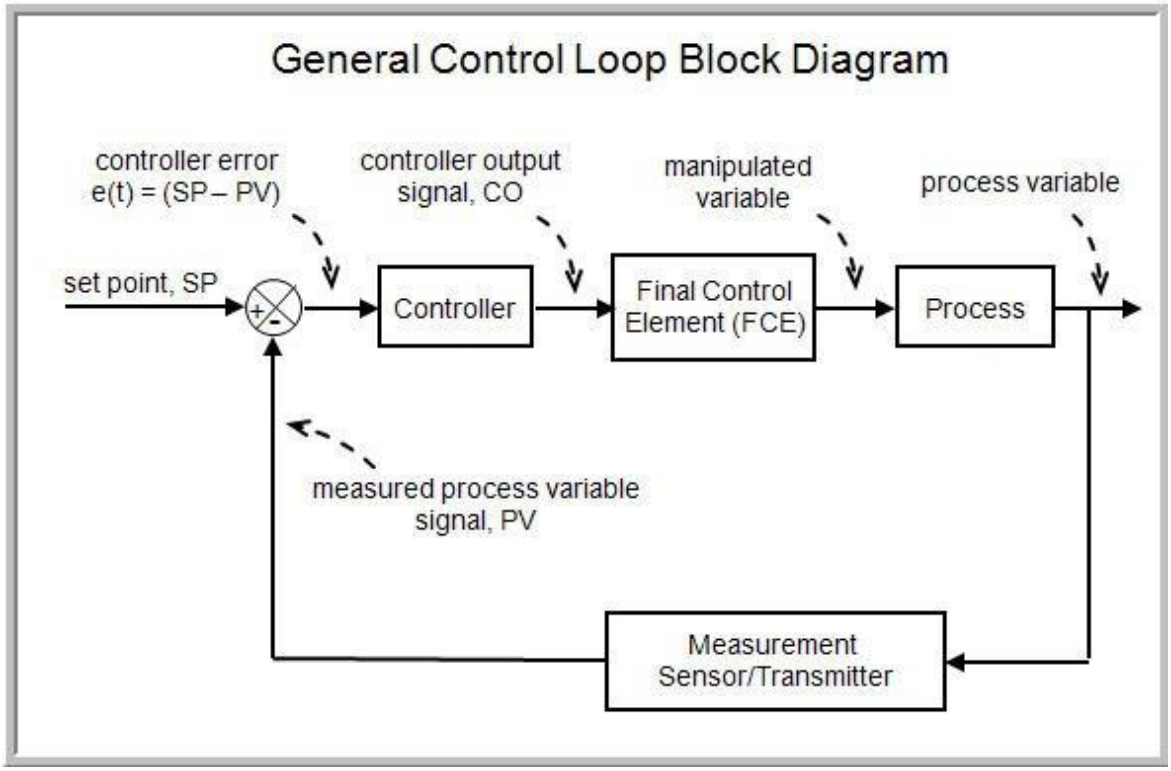


Fig (3-14)



Al-Balqa' Applied University

Faculty of Engineering Technology

Mechatronics Engineering Department

Pneumatic and hydraulic Drive Lab

Ex 4

Control pneumatic system using PLC with Directional control valves, and electrical limit switches

Student's Name:

Supervisor:

Date:

❖ Objective:

1. The purpose of this experiment is to learn how to control a double acting cylinder using PLC.
2. In this part student will assemble, test and operate the drive circuit in order to outstroke and in stroke the cylinder after pressing the push button.

❖ Necessary Equipment:

1. Two double acting pneumatic cylinders.
2. Two 5/2 directional control valves (impulse type).
3. Four electrical limit switches.
4. Two electrical push-button.
5. PLC.

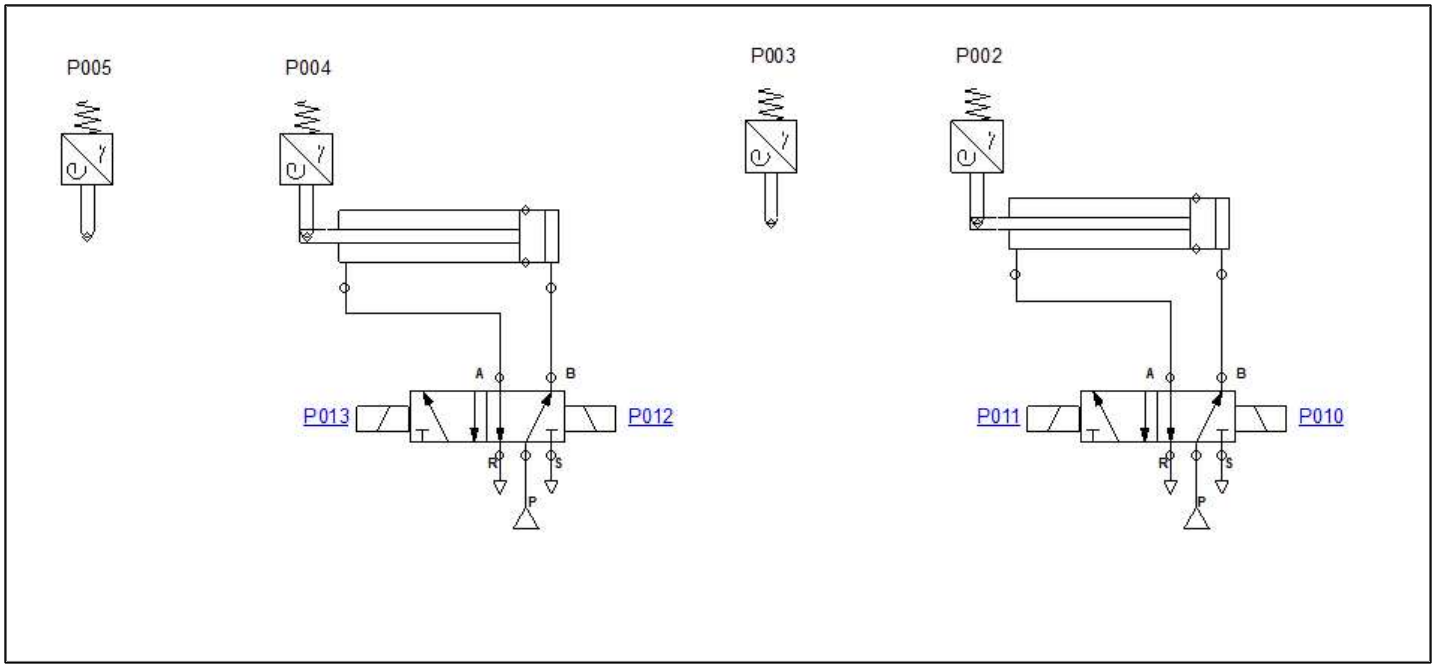


Fig (4-1)

❖ LAB ACTIVITY:

1. **Activity (1)**: Draw the pneumatic drive circuit using standard symbols in order to realize the following sequence.

A+	B+	A-	B-

2. **Activity (2)**: Draw the block diagram and signal flow graph as in the Fig (4-2)

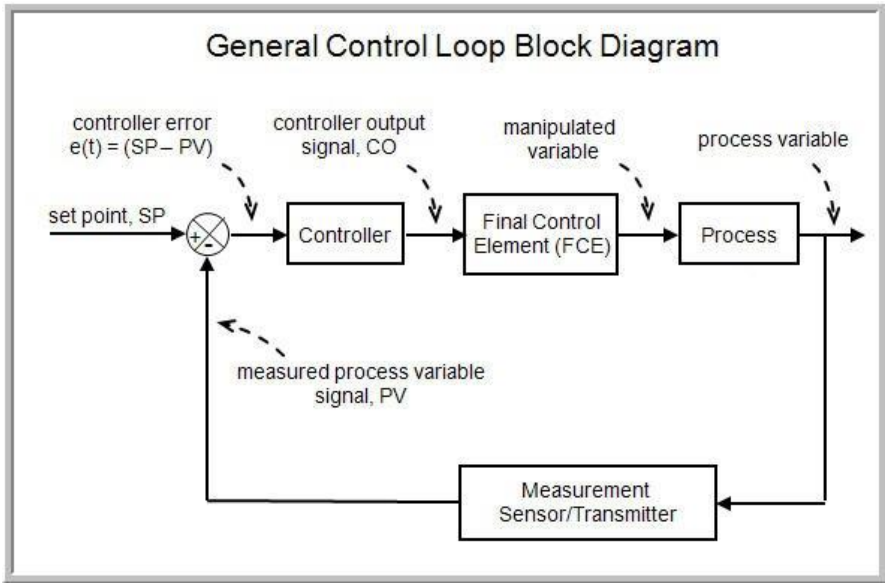


Fig (4-2)

❖ HOME ASSIGNMENTS:

1. Simulate the circuit using Automation studio.
2. Consider the following sequence.

A+	B+	B-	A-

   A. Build a circuit within order to realize the following sequence.
   B. Write a PLC program.

3. What happens if the push button is continuously pressed?

Manual Programmer PLC ELC 2001 PL

- **Prepare your PLC for programming:**

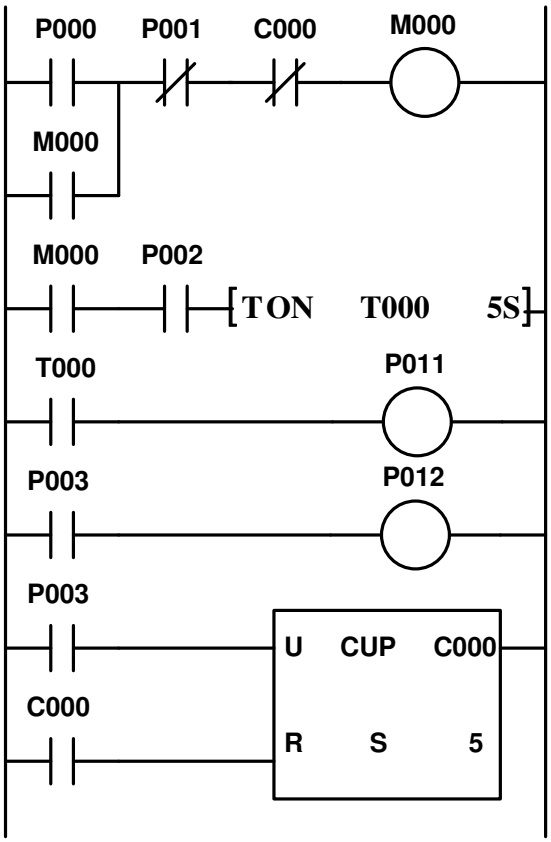
1. Run power supply.

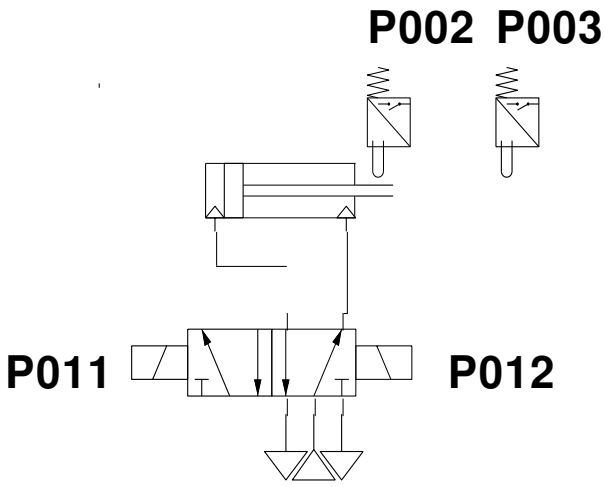
2. Before edition your program the programmer must be in program mode:
To do this press Key "1", to select "Programmer" and then press "ENTER".

3. You have to clear the PLC memory and the program memory:
To do this press key "SHFT + CLR" and then press key "SHFT + DELT".

4. Edit your program:

a. Enter your program, for example:





LOAD	**P000**	**(ENT)**
OR	**M000**	**(ENT)**
ANDNOT	**P001**	**(ENT)**
AND NOT	**C000**	**(ENT)**
OUT	**M000**	**(ENT)**
Load	**M000**	**(ENT)**
AND	**P002**	**(ENT)**
TMR	**T000**	**(ENT)**
DATA	**00050**	**(ENT)**
LOAD	**T000**	**(ENT)**
OUT	**P011**	**(ENT)**
LOAD	**P003**	**(ENT)**
LOAD	**C000**	**(ENT)**
CNT	**C000**	**(ENT)**
DATA	**0005**	**(ENT)**

b. To end your program:

Press key "FUN" and then press key "1" and then press key "ENTER".

5. Load your program to the PLC:
To do this press key "RUN".

/

Al-Balqa' Applied University

Faculty of Engineering Technology

Mechatronics Engineering Department

Process control Lab

Ex. 5

Student's Name:

Supervisor:

Date:

❖ Objective:

Definition the student in a way The Ziegler–Nichols tuning method

❖ Introduction:

The Ziegler–Nichols tuning method is a heuristic method of tuning a PID controller. It was developed by “John G. Ziegler” and “Nathaniel B. Nichols”. It is performed by setting the, I (integral) and D (derivative) gains to zero. The "P" (proportional) gain, K_P is then increased (from zero) until it reaches the ultimate gain K_U at which the output of the control loop has stable and consistent oscillations. K_U and the oscillation period T_U are used to set the P, I, and D gains depending on the type of controller used.

- What is the Ziegler-Nichols Rule?

The Ziegler-Nichols rule is a heuristic PID tuning rule that attempts to produce good values for the three PID gain parameters:

1. Kp - the controller path gain
2. Ti - the controller's integrator time constant
3. Td - the controller's derivative time constant

Given two measured feedback loop parameters derived from measurements:

1. the period Tu of the oscillation frequency at the stability limit
2. the gain margin Ku for loop stability

With the goal of achieving good regulation (disturbance rejection)

- When Does the Ziegler-Nichols Rule Work?

Tuning rules work quite well when you have an analog controller, a system that is linear, monotonic, and sluggish, and a response that is dominated by a single-pole exponential "lag" or something that acts a lot like one.

❖ **Necessary Equipment:**

1. Computer.
2. le cableUniversal digital controller (Consists of):
3. Supply voltage , +15 volt:
4. Maximum current consumption here is 600 mA.
5. Connection rail for 5 v supply voltage:
6. This rail is used for through connection of a 5 v supply that may be available. The 5 v is not required for this controller panel.
7. Supply voltage, ground rail:
8. Socket for power supply: earth (0v) connection.
9. Supply voltage,-15volt:
10. Socket for connecting the negative supply voltage
11. Maximum current consumption here is 60 mA.
12. Voltage output (v1,v2):
13. 2 potentiometers for adjusting the programmable reference voltages, that are available at 4mm safety socket.
14. LC-display:
15. With graphic capabilities, LC-display with background illumination resolution 128*64 dots
16. Serial interface RS232:
17. This socket must be connected to the serial interface when operating with a PC. Use a standard 9-po.

- Pushbutton for switching the controller ON or OFF (RUN/STOP):
  Operation of the controller is indicated by a LED integrated in the pushbutton.
- Operating keypad:
  Six keys for adjustments and operating the controller
- Analog inputs(W,X):

Four analog inputs with measurement range of ±10v, ±5v or ±2-5v, adjustable by software Optional auto ranging facility 10 bit resolution. Input impedance 10 KΩ Maximum permissible input voltage (± 15v).

- Analog outputs (y):

  4mm safety sockets. Maximum output voltage, ±10v. Status indication via LED (overdrive) .maximum output current, 4˜10 mA.

❖ Procedure:

1. Turn on the computer.
2. open program "PID Controller" from desktop
3. Press this icon  from the toolbar.
4. From “sittings” Press “controller parameter”, from the list "control input" press "manual".
5. From “chart” press “properties”.
   A. from the list “signals” Select the following signals:
      - “Control input” Choose a specific color.
      - “Controlled variable” Choose a specific color.
      - “Upper switching point” &“lower switching point” Choose a specific color.
   B. from the list “scale” Select the following:
      - “Maximum” writ “200”.
      - “Division” writ “10”.

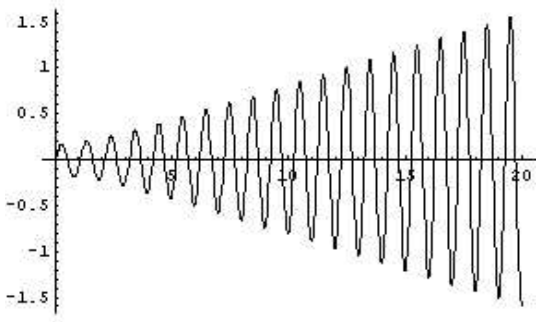
ZIGLER-NICHOLS CLOSED LOOP TUNNING METHOD

1. The introduction of the proportional (gain) Kp into the system.
2. Remove the differential and integral gain from the system.

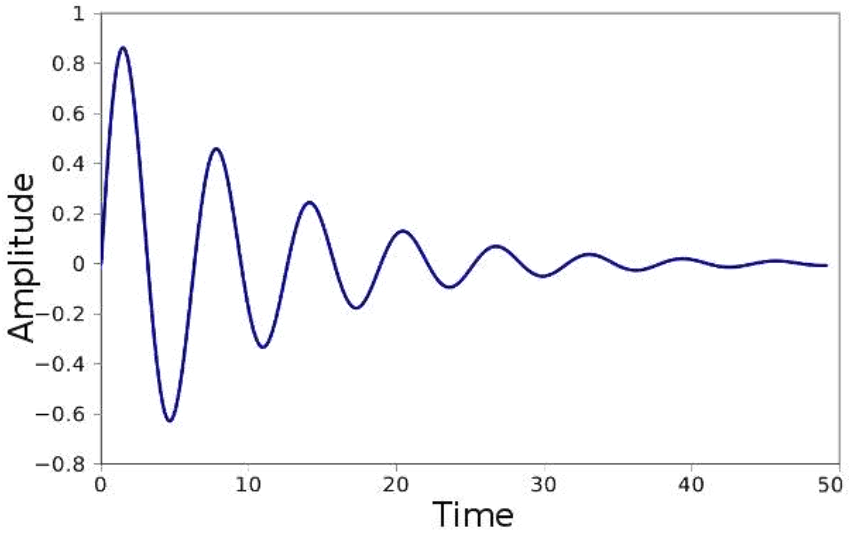
$Ki = \frac{1}{Ti}$ Ki = zero Kd= Td=zero

Ti = ∞

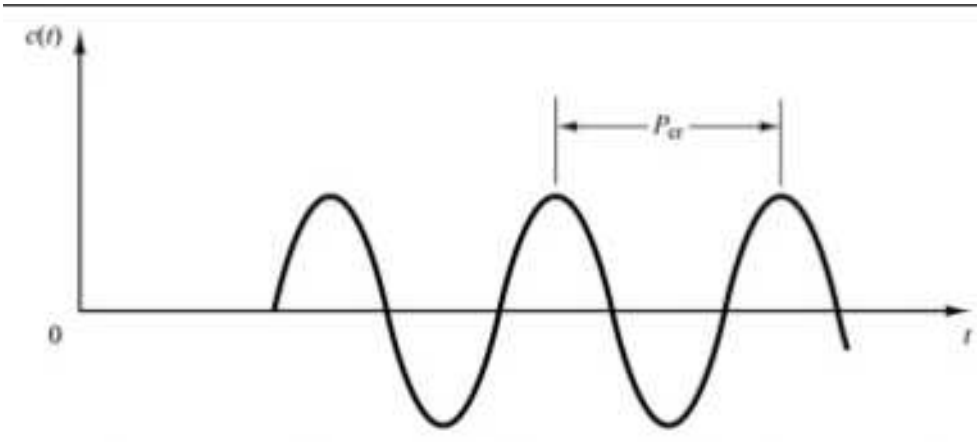
3. Disturbance the system by making a small change to (S.P).
4. Increase the proportional gain Kp.
   - If an undamped signal is obtained as shown in the figure below, this means that the Kp value entered is large and should be reduced.



   - If a damped signal is obtained over time, as in the figure below, the Kp value entered should be increased



- We increase and decrease the Kp we get the “Ultimate proportional gain” Kpu and “Ultimate period” Pu by obtaining the oscillating and stable signal as if in the phase of the on-off controller as in the figure below



5. Here we calculate the Ultimate proportional gain and Ultimate period
   Kpu= Ultimate proportional gain (الكسب التناسبي الاقصى)
   Pu= Ultimate period (الدورة القصوى)
   A. Proportional controller – P mode.

$K_P = 0.5K_{PU}$

B. Proportional-Integral controller– PI mode.

$K_P = 0.45\ K_{PU}$ $\qquad$ $Ti = \frac{Pu}{1.2}$

C. Proportional-Derivati ve controller– PD mode.

$K_P = 0.6\ Kpu$ $\qquad$ $K_d = Td = \frac{Pu}{8}$

D. Proportional-Integral-Derivative controller– PID mode.

$Kp = 0.6\ Kpu$ $\qquad$ $Ti = \frac{Pu}{2}$ $\qquad$ $Kd = Td = \frac{Pu}{8}$

❖ **HOME ASSIGNMENTS:**

Draw the block diagram and signal flow graph as in the Fig (5-1)

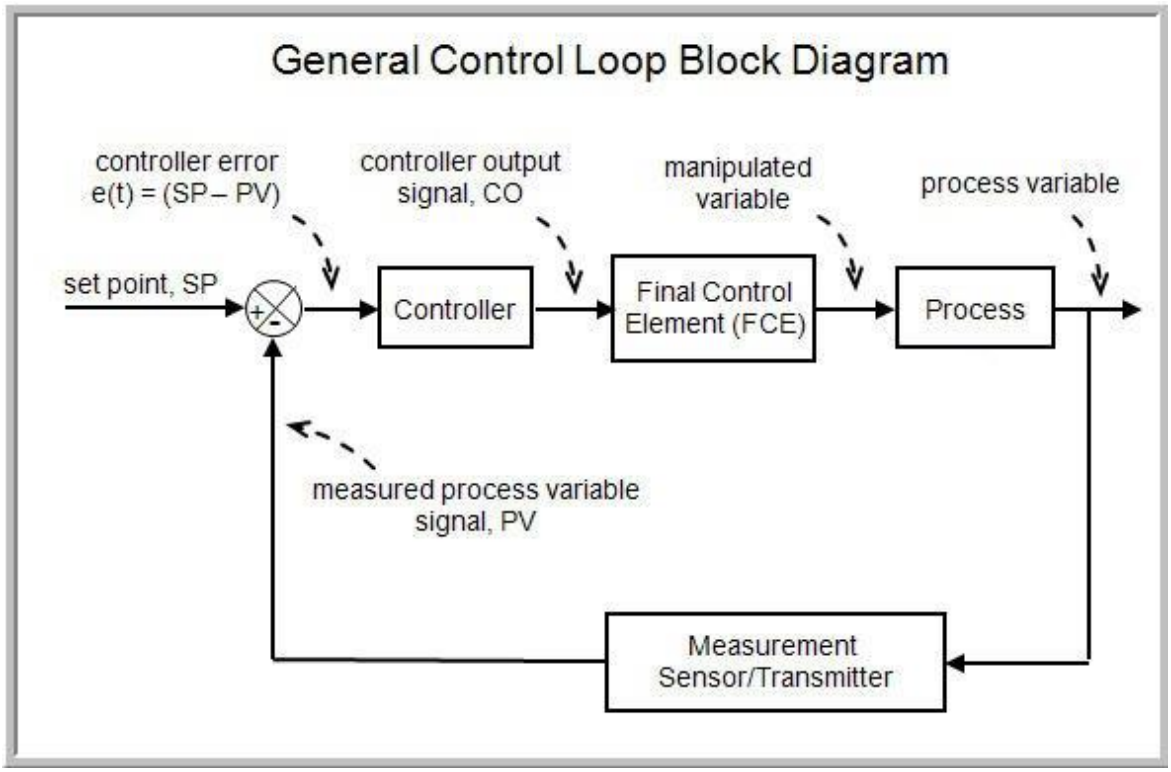

General Control Loop Block Diagram
controller error
e(t) = (SP – PV)
controller output
signal, CO
manipulated
variable
process variable
set point, SP
+
-
Controller
Final Control
Element (FCE)
Process
measured process variable
signal, PV
Measurement
Sensor/Transmitter

Fig (5-1)

W

Al-Balqa' Applied University

Faculty of Engineering Technology

Mechatronics Engineering Department

Process control Lab

Ex. 6

PID controllers of the Flow

(UCP-P)

Student's Name:

Supervisor:

Date:

1. Monitoring the optimization process of a PID controller for a given process.

❖ Objective:

2. with the most commonly used methods for optimizing the variables of a PID controller from the characterization of the process.Become familiar

❖ Necessary Equipment:

1. Control valve.
2. Pneumatic valve.
3. Relative pressure manometer.
4. Absolute pressure sensor.
5. Storage tank.
6. Inlet/outlet valve.
7. Diaphragm.
8. Exhaust valve.
9. Pointer flow measurer.
10. Differential pressure measurer.
11. I/P converter.
12. Control interface.

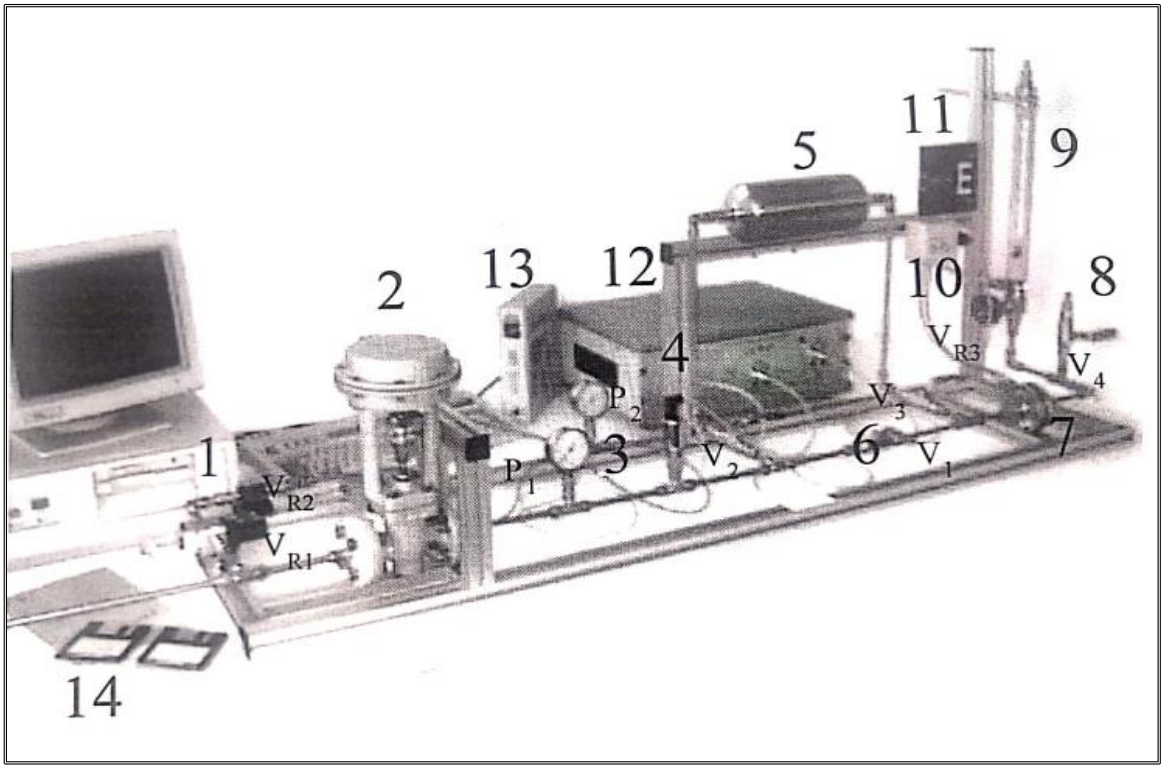


Fig (6-1)

❖ Produce:

1. Valves V_2, V_3 must be closed and V_{R-3} must be 30% regulated. Valves V_1 must be open and V_4 must be closed.
2. Open Pc and system software.
3. Adjust the "consign value "on "50 bar".
4. Increase the proportional band from 10% to 20 % to 30 % to 50% until the system stable (P range 01% ~ 100%)
5. Add the integral range until the offset become zero (I range .1 ~ 10)
6. Add the differential range until the system is accelerated (D range .01 ~ .5).

❖ HOME ASSIGNMENTS:

Draw the block diagram and signal flow graph as in the Fig (6-2)

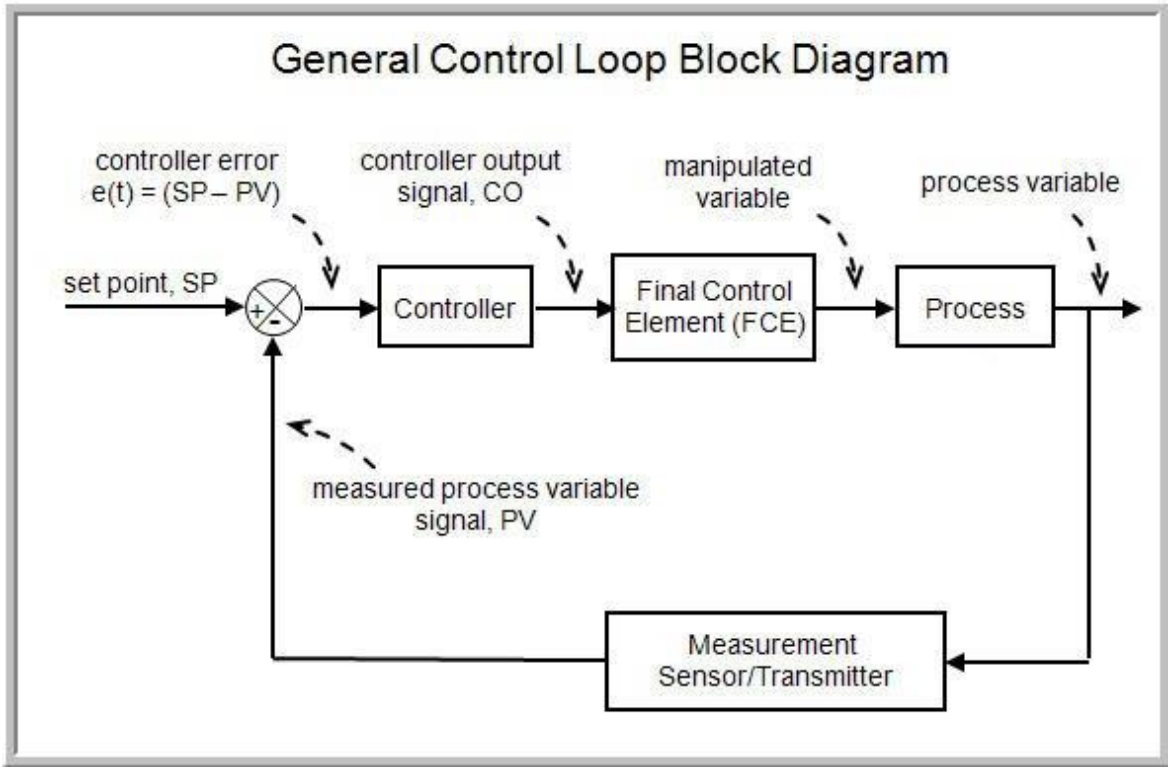


Fig (6-2)